Neuroscience

Rats emit aversive 44-kHz ultrasonic vocalizations during prolonged Pavlovian fear conditioning

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Abstract

Rats are believed to communicate their emotional state by emitting two distinct types of ultrasonic vocalizations. The first is long "22-kHz" vocalizations (>300 ms, <32 kHz) with constant frequency, signaling aversive states and the second, is short "50-kHz" calls (<150 ms, >32 kHz), often frequency-modulated, in appetitive situations. Here we are first to describe a new-type of aversive vocalizations emitted at a higher pitch by Wistar and spontaneously hypertensive rats (SHR) in an intensified aversive state – prolonged fear conditioning. These calls, which we named "44-kHz" vocalizations, are long (>150 ms), generally at a constant frequency (usually within 35-50 kHz range) and have an overall spectrographic image similar to 22-kHz calls. Some 44-kHz vocalizations are comprised of both 22-kHz-like and 44-kHz-like elements. Furthermore, two separate clustering methods confirmed that these 44-kHz calls are distinct from other vocalizations. We observed 44-kHz calls to be associated with freezing behavior during fear conditioning, during which they constituted up to 19.4% of all calls. We also show that some of rats' responses to the playback of 44-kHz calls were more akin to that of aversive calls, e.g., heart rate changes, whereas other responses were at an intermediate level between aversive and appetitive calls. Our results suggest that rats have a wider vocal repertoire than previously believed, and current definitions of major call types may require reevaluation. We hope that future investigations of 44-kHz calls in rat models of human diseases will contribute to expanding our understanding and therapeutic strategies related to human psychiatric conditions.

eLife assessment

This **useful** study investigated the appearance of a "new-type" ultrasonic vocalization around 44 kHz that occurs in response to prolonged fear conditioning in rats. While the descriptive approach applied may be of interest to some researchers, evidence in support of the conclusions is **incomplete**.

eLife

Introduction

Charles Darwin wrote: “That the pitch of the voice bears some relation to certain states of feeling is tolerably clear” (Darwin, 1872). This has also been tolerably clearly observed and widely described for ultrasonic vocalizations of rats (Brudzynski, 2019, Brudzynski, 2021, Simola and Granon, 2019) which emit low-pitched aversive calls and high-pitched appetitive calls. The former are “22-kHz” vocalizations (**Figs 1A**, **2A**), with 18 to 32 kHz frequency range, monotonous and long, usually >300 ms, and are uttered in distress (Brudzynski, 2013, Brudzynski, 2019, Brudzynski, 2021, Simola and Granon, 2019). The latter are “50-kHz” vocalizations (**Fig. 1C**), are relatively short (10-150 ms), frequency-modulated, usually within 35-80 kHz, and they signal appetitive and rewarding states (Simola and Granon, 2019, Brudzynski, 2013, Brudzynski, 2019, Brudzynski, 2021). Therefore, these two types of calls communicate the animal’s emotional state to their social group (Brudzynski, 2013). Low-pitch (<32 kHz), short (<300 ms; **Fig. 1B**) calls, assumed to also express a negative aversive state, have been described but their role is not clearly established (Brudzynski, 2013). Notably, high-pitch (>32 kHz), long and monotonous ultrasonic vocalizations have not yet been described. Here we show these unmodulated rat vocalizations with peak frequency[1] at about 44 kHz (**Figs 1B**, **1E**, **2B**), emitted in aversive experimental situations, especially in prolonged fear conditioning.

Results

New calls are high, long, unmodulated

In three separate experiments (all summarized in **Tab. 1**/Exp.1-3, see Methods), i.e., one with trace-fear-conditioning (**Tab. 1**/Exp. 1) and two with delay-fear-conditioning (**Tab. 1**/Exp. 2-3), one of which has already been described (**Tab. 1**/Exp. 2, Olszyński et al., 2021, Olszyński et al., 2022), 53 of all 84 conditioned Wistar rats (**Tab. 1**/Exp. 1-3/#2,4,6-8,13, **Figs 1B**, **1E**, **1S1BC**) displayed vocalizations that were high-pitched, i.e., in the range of 50-kHz calls, but long and monotonous (**Fig. 2B**). These vocalizations, e.g., top-right group in **Figs 1B** and 1S1C, were outside the defined range (Brudzynski, 2019, Brudzynski, 2021, Simola and Granon, 2019) for both 50-kHz (bottom-right group in **Figs 1C**, **1S1A-C**) and 22-kHz calls (top-left group in **Figs 1A**, **1B**, **1S1A-C**). These vocalizations were also observed in a different rat strain acquired from a different breeding colony, i.e., spontaneously hypertensive rats (SHR) (Okamoto and Aoki, 1963), also trained in delay fear conditioning (**Tab. 1**/Exp. 2/#10-12; Olszyński et al., 2022). Six of the 49 conditioned SHR displayed high-pitch, long, monotonous vocalizations (e.g., **Fig. 2S1G**); moreover, we observed more of these vocalizations in Wistar rats compared to SHR (**Tab. 1**/Exp. 2/#6-8,10-12) in both training, $p < 0.0001$, and test sessions, $p = 0.0030$, Mann-Whitney.

Overall, we analyzed 140,149 vocalizations from all fear conditioning experiments (**Tab. 1**/Exp. 1-3/#1-13, n = 218) and through trial-and-error, we set **new criteria**, namely **peak frequency of >32 kHz and >150 ms duration** to define the new-type calls. We manually verified the results on the spectrogram using these parameters and only 308 calls (0.2%) were incorrectly assigned (i.e., exceptionally long 50-kHz vocalizations misplaced in the new-type group or borderline-short vocalizations of the new-type misplaced to 50-kHz calls). Hence the new parameters correctly assigned 99.8% of cases and are thus effective to distinguish the new-type calls in an automated fashion. Finally, 10,445 new-type calls were identified, which constituted 7.5% of the total calls during fear conditioning experiments (**Tab. 1**/Exp. 1-3; comp. **Fig. 1G**). These vocalizations have a peak frequency range from 32.2 to 51.5 kHz (95% of cases) with an average peak frequency

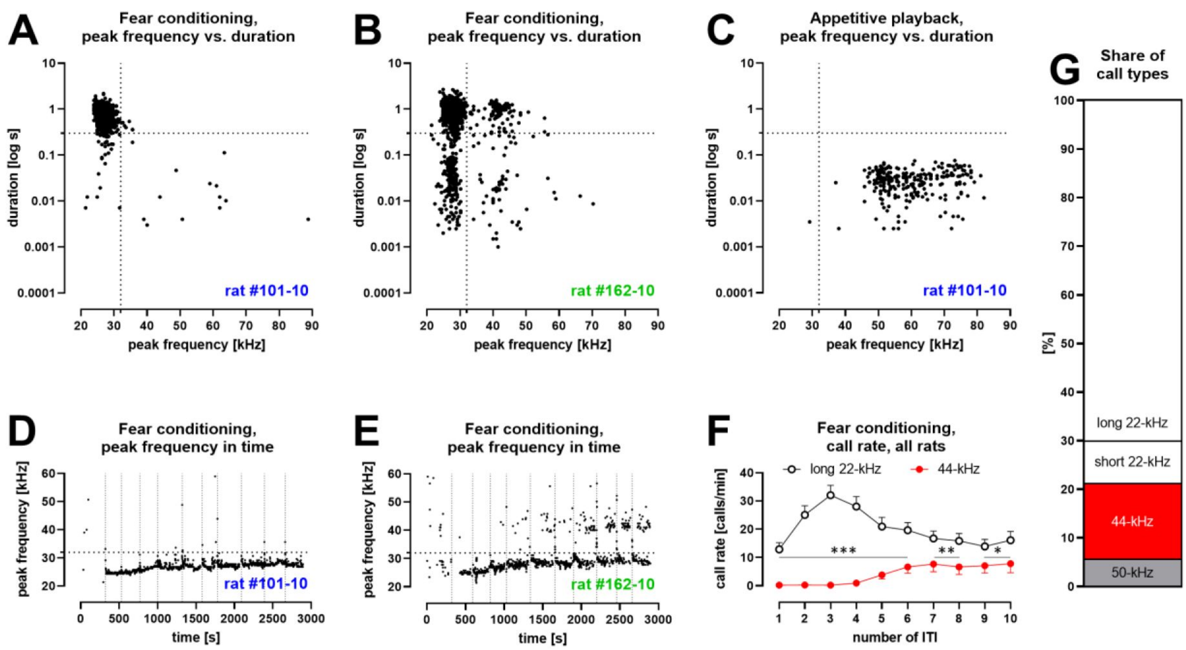

Fig. 1.

Characteristics of vocalizations emitted by Wistar rats during fear conditioning with ten aversive foot-shocks

(**Tab. 1** / Exp. 1-3/#2,4,8,13; n = 46)**. A** – some rats produced aversive 22-kHz vocalizations with typical features, i.e., constant-frequency of <32 kHz, >300 ms duration – both values marked as dotted lines); example emission from one rat. **B** – some rats produced 44-kHz vocalizations with constant frequency of >32 kHz and long duration (>150 ms); example emission from one rat. **C** – rats which emitted aversive vocalizations during fear session, produced 50-kHz vocalizations during appetitive playback session the following day (full data published in Olszyński et al., 2021); representative data from same rat in **A**. **D** – the onset of long 22-kHz alarm calls typically occurred after first shock stimulus (vertical dotted lines mark time of shock deliveries in **DE**); note the gradual rise in peak frequency[a], not exceeding 32 kHz (horizontal dotted line in **DE**); data from the same rat as **AC**. **E** – in rats that emitted 44-kHz calls, the onset was usually delayed to after several foot-shocks; note the gradual rise in peak frequency of both long 22-kHz and 44-kHz vocalizations throughout training (comp. **Fig. 1S2CD**); data from same rat in **B**). **F** – call rate of long 22-kHz calls was higher than 44-kHz calls (*p < 0.05, **p < 0.01, ***p < 0.001) and with different time-course – maximum number of 22-kHz calls at ITI-3 (higher than ITI-1, 2, 5-10; <0.0001–0.0005 p levels); and higher number of 44-kHz calls at ITI-5-10, i.e., 6.6 ± 2.3 vs. ITI-1-4, i.e., 0.4 ± 0.2; p < 0.0001; all Wilcoxon); numbers of ITI (inter-trial-intervals) correspond to the numbers of previous foot-shocks, values are means ± SEM. **G** – long 44-kHz vocalizations had a higher incidence rate (15.5%) than short 22-kHz (8.8%) and 50-kHz calls (5.6%); values are calculated for sum of all vocalizations obtained during entire training sessions (there were fewer 50-kHz calls, i.e., 3.7%, when vocalizations prior to the first shock were not included). **A-E**: dots reflect specified single rat values. **FG**: n = 46, other results from these rats are previously published (Olszyński et al., 2021, Olszyński et al., 2022).

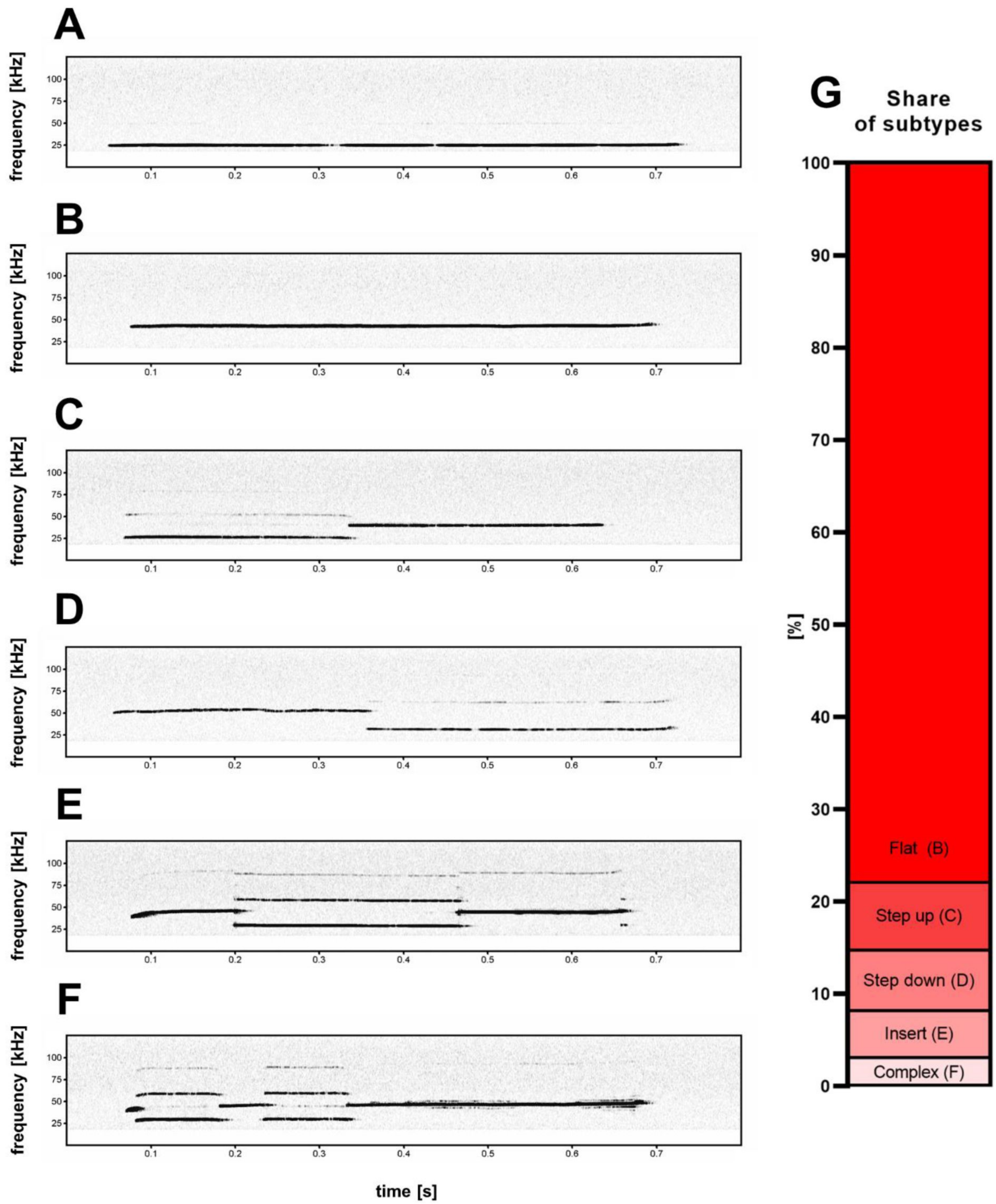

Fig. 2.

Five subtypes (B-F) of high frequency 44-kHz aversive vocalizations.

A – standard aversive 22-kHz vocalization with peak frequency <32 kHz (peak frequency = 24.4 kHz). 44-kHz aversive vocalization subtypes: **B** – flat (constant frequency call; peak frequency = 42.4 kHz), **C** – step up (peak frequency = 39.5 kHz), **D** – step down (peak frequency = 52.2 kHz), **E** – insert (peak frequency = 38.5 kHz), **F** – complex (peak frequency = 46.3 kHz). **G** – percentage share of 44-kHz call-subtypes in all cases of detected 44-kHz vocalizations.

<table>
<tr><th>#</th><th>Exp.</th><th>FC type</th><th>strain</th><th># of trials (shocks)</th><th># of rats (n)</th><th>housing</th><th>trans-mitters</th><th>age in weeks</th><th>description/history</th></tr>
<tr><td>1</td><td rowspan="4">1</td><td rowspan="4">trace</td><td rowspan="4">Wistar</td><td>0*</td><td>7</td><td rowspan="2">single</td><td rowspan="4">+</td><td rowspan="4">13</td><td rowspan="4">Rats that undergone FC after playback experiments (published in Olszyński et al., 2020)</td></tr>
<tr><td>2</td><td>10</td><td>7</td></tr>
<tr><td>3</td><td>0*</td><td>10</td><td rowspan="2">paired</td></tr>
<tr><td>4</td><td>10</td><td>10</td></tr>
<tr><td>5</td><td rowspan="8">2</td><td rowspan="8">delay</td><td rowspan="4">Wistar</td><td>0*</td><td>37</td><td rowspan="8">paired</td><td rowspan="8">+</td><td rowspan="8">12</td><td rowspan="8">FC experiments which were described before in detail (Olszyński et al., 2021, Olszyński et al., 2022)</td></tr>
<tr><td>6</td><td>1</td><td>16</td></tr>
<tr><td>7</td><td>6</td><td>22</td></tr>
<tr><td>8</td><td>10</td><td>19</td></tr>
<tr><td>9</td><td rowspan="4">SHR</td><td>0*</td><td>31</td></tr>
<tr><td>10</td><td>1</td><td>17</td></tr>
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<tr><td>12</td><td>10</td><td>15</td></tr>
<tr><td>13</td><td>3</td><td>delay</td><td>Wistar</td><td>10</td><td>10</td><td>paired</td><td>-</td><td>12</td><td>New FC experiment</td></tr>
</table>

Table 1.

All fear conditioning (FC) experiments described in the text.

* – control groups

of 42.1 kHz, and they exhibited 43.8 kHz peak frequency at the cluster center in a DBSCAN analysis (**Fig. 3A**). In line with the accepted nomenclature convention, underlining the relationship with 22-kHz vocalizations, we christened this new-type of ultrasonic calls as “**44-kHz vocalizations**”.

44-kHz calls in long aversive stimulation

We found 44-kHz vocalizations especially in rats which received multiple electric shocks. When we analyzed all Wistar rats that had undergone 10 trials of fear conditioning (**Tab. 1**/Exp. 1-3/#2,4,8,13; n = 46), these vocalizations were less frequent following the first trial (1.2 ± 0.4% of all calls), and increased in subsequent trials, particularly after the 5th (8.8 ± 2.8%), through the 9th (19.4 ± 5.5%, the highest value), to the 10th (15.5 ± 4.9%) trials, where 44-kHz calls gradually replaced 22-kHz vocalizations in some rats (**Fig. 1F**, 1S2AB, Video 1; comp **Fig. 1D** vs. 1E). Please note, majority of the 22-kHz calls were emitted after the 3rd shock, i.e., during the 3rd ITI (inter-trial-interval), while 44-kHz vocalizations were emitted in the second part of the training, i.e., 5th to 10th ITI (**Fig. 1F**, comp. **Fig. 1S2AB**). From this group of rats (n = 46), n = 41 (89.1%) emitted long 22-kHz calls, and 32 of them (69.6%) emitted 44-kHz calls, i.e., every animal that produced 44-kHz calls also emitted long 22-kHz calls (**Fig. 1S2AB**). The prevalence of 44-kHz calls varied greatly among individual rats, such that for n = 3 rats, 44-kHz vocalizations accounted for >95% of all calls during at least one ITI (e.g., 140 of total 142, 222 of 231, and 263 of 265 tallied 44-kHz calls), and in n = 9 rats, 44-kHz vocalizations constituted >50% of calls in more than one ITI. The prevalence of 44-kHz calls in all experimental conditions analyzed in all animal groups is shown in **Fig. 1S**3.

Notably, there were more 44-kHz vocalizations during fear conditioning training than testing in all fear-conditioned Wistar rats (**Tab. 1**/Exp. 1-3/#2,4,6-8,13; n = 84; 3.63 ± 0.99 vs. 0.23 ± 0.13 calls/min; p < 0.0001; Wilcoxon).

In a recent publication during this paper’s review process, Gonzalez-Palomares et al. (2023), inspired by our current findings, investigated and reported 44-kHz vocalizations following prolonged (10-trial procedure) odor fear conditioning. These calls were observed predominantly during the late ITI, i.e., 8th-10th ITI (Gonzalez-Palomares et al., 2023; Fig. S4C; please note 4th-7th ITI were not investigated) after the shock presentations (Fig. S4B therein), which complement our results.

Changes in frequency, duration, and mean power of long aversive calls during conditioning

Analyzing Wistar rats that undergone 10 trials of fear conditioning (**Tab. 1**/Exp. 1-3/#2,4,8,13; n = 46), we also observed the frequencies of 22-kHz calls to gradually rise throughout fear conditioning training, i.e., during subsequent ITI – from 24.5 ± 0.1 to 27.9 ± 0.4 kHz (**Figs 1DE**, **1S2C**; p < 0.0001, Friedman, p = 0.0039, Wilcoxon). The frequency levels of 44-kHz vocalizations also appeared to rise – from 37.8 ± 2.1 to 39.6 ± 1.3 kHz (**Figs 1E**, **1S2C**) but we were unable to statistically demonstrate it (p = 0.0155, Friedman, p = 0.0977, Wilcoxon).

There was a shortening of long 22-kHz calls during the first four ITI from 969.6 ± 43.1 ms to 794.6 ± 39.8 ms (p < 0.0001, Friedman; p < 0.0001, Wilcoxon, **Fig. 1S2D**), while 44-kHz vocalizations were longest during the 4th ITI (the time of their substantial appearance, comp. **Fig. 1F**), i.e., 775.0 ± 135.7 ms, and shortened over subsequent ITI (619.6 ± 58.1 ms for the 10th ITI, **Fig. 1S2D**, p = 0.0227, Friedman; p = 0.0234, Wilcoxon).

Finally, the sound mean power of 44-kHz vocalizations appeared to remain stable throughout the 10-trial sessions, while during the first half of the training, i.e., 1st-5th ITI, 22-kHz calls were not only significantly more frequent but also louder than during the second half, i.e., 6th-10th ITI (p < 0.0001, Wilcoxon). Consequently, long 22-kHz calls appeared louder than 44-kHz calls (p = 0.0397-

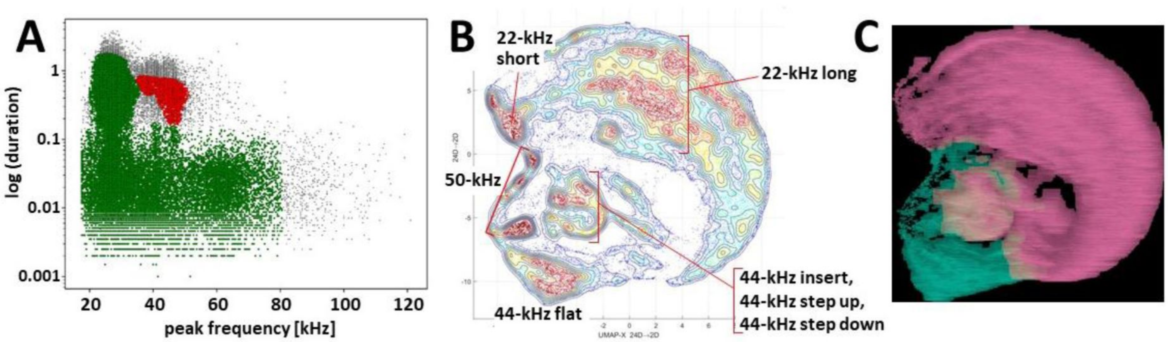

Fig. 3.

Clustering of ultrasonic vocalizations from fear conditioning sessions using two independent methods.

A – DBSCAN algorithm (ε = 0.14) clustering of vocalizations from all fear conditioning experiments (**Tab. 1**/Exp. 1-3/#1-13, n = 218), silhouette coefficient = 0.198, two clusters emerge, cluster of green dots n = 77,243 (due to high generality of cluster average peak frequency and duration deemed redundant), cluster of red dots n = 5,646 (average peak frequency = 43,826.6 Hz, average duration = 0.524 s), some calls were not assigned to any cluster, i.e., outlier vocalizations, black dots, n = 4,139. **BC** – clustering by k-means algorithm and visualization of calls emitted by selected rats, i.e., with >30 of 44-kHz vocalizations, during trace and delay fear conditioning training (n = 26, selected from **Tab. 1**/Exp. 1-3/#2,4,7,8,11-13), total number of calls n = 40,084. **B** – topological plot of ultrasonic calls using UMAP embedding, particular agglomerations of calls labeled with their type or subtype. **C** – spectrogram images from DeepSqueak software superposed over plot B, colors denote clusters from unsupervised clustering, number of clusters set using elbow optimization (max number = 4), two clusters emerge; see also **Fig. 3S1**.

0.0038, Mann-Whitney). However, in the second half of the session, this difference dissipated due to the diminishing amplitude of 22-kHz vocalizations (p = 0.0083, Friedman; p = 0.0046, Wilcoxon), while the amplitude of 44-kHz calls remained stable (p = 0.0663, Friedman; p = 0.2661, Wilcoxon; 6th ITI through 10th ITI for both; **Fig. 1S2E**). After adjusting for angle-dependent hardware attenuation (see Methods, Sound mean power), the situation reversed (**Fig. 1S2F**). Both long 22-kHz and 44-kHz vocalizations showed similar amplitude levels during the first half of the fear conditioning session, while during the 6th-10th ITI, 44-kHz calls were significantly louder than long 22-kHz calls (p = 0.0007-0.0097, Mann-Whitney).

44-kHz calls linked to freezing

We investigated the freezing behavior of all Wistar rats emitting 44-kHz vocalizations during 10 trials of fear conditioning (**Tab. 1**/Exp. 1-3/#2,4,8,13; n = 46). The training sessions were divided into 10-s-long time bins, from which we analyzed only the bins that had exclusively long 22-kHz or 44-kHz calls. For comparison, we also measured the freezing levels during the first 5 min of the trial (baseline freezing levels before any foot-shocks) as well as the bins in which animals did not vocalize (from the period after the 1st shock to the end of the session). Of the n = 46 rats analyzed, n = 41 emitted 22-kHz vocalizations, from which n = 32 also emitted 44-kHz vocalizations, from which only n = 21 were determined to have both – 10-s-long bins of 22-kHz calls only and 44-kHz calls only (**Tab. 2**A). Freezing during the bins of 22-kHz calls only (p < 0.0001, for both groups) and during 44-kHz calls only bins (p = 0.0003) was higher than during the first 5 min baseline freezing levels of the session. Also, the freezing associated with emissions of 44-kHz calls only was higher than during bins with no ultrasonic vocalizations (p = 0.0353), and it was also 9.9 percentage points higher than during time bins with only long 22-kHz vocalizations, but the difference was not significant (p = 0.1907; all Wilcoxon).

To further investigate this potential difference, we measured freezing during the emission of randomly selected single 44-kHz and 22-kHz vocalizations. The minimal freezing behavior detection window was reduced to compensate for the higher resolution of the measurements (3, 5, 10, or 15 video frames were used). There was no difference in freezing during the emission of 44-kHz vs. 22-kHz vocalizations for ≥150-ms-long calls (3 frames, p = 0.2054) and for ≥500-ms-long calls (5 frames, p = 0.2404; 10 frames, p = 0.4498; 15 frames, p = 0.7776; all Wilcoxon, **Tab. 2**B).

44-kHz calls sorted into five subtypes

While the majority of 44-kHz vocalizations were of continuous unmodulated frequency (**Fig. 2B**), some comprised additional elements. Based on the composition of individual call elements and their relation to each other, we manually sorted the calls into five categories (**Fig. 2B-F**). If the start (*prefix*) or end (*suffix*) portion of a call was less than 1/5th the length of the following or previous element, this portion of the call was not considered in its categorization into the five subtypes. The names and descriptions of the five subtypes are: **flat** – single element with near constant frequency and little to no interruptions to the sound continuity on the spectrogram; **step up** – two elements with an instantaneous frequency jump, where the first element is of lower frequency; **step down** – two elements with an instantaneous frequency jump, where the first element is of higher frequency; **insert** – three elements with an instantaneous frequency change, where the middle element is of different frequency; **complex** – more than three elements with instantaneous frequency changes.

44-kHz and 22-kHz calls closely related

44-kHz were emitted in aversive behavioral situations – as 22-kHz calls are observed (Antoniadis and McDonald, 1999, Dupin et al., 2019, Taylor et al., 2017). Both types of calls are long (usually >300 ms) and frequency-unmodulated. Some of the elements constituting such as step up; step down; insert and complex 44-kHz vocalizations (**Fig. 2C-F**) were at a lower frequency –

A.				**10-s-long time intervals**; analyzed with 30 frames
group of rats analyzed	freezing [%]			
	first 5 min	10-s bins with:		
		w/o calls	22-kHz calls only	44-kHz calls only
rats with long 22-kHz calls (n = 41)	6.3 ± 2.3	39.7 ± 3.5	47.8 ± 3.3 $^{***,\#}$	NA
rats with long 22-kHz calls and 44-kHz calls (n = 21)	9.1 ± 4.2	33.7 ± 4.9	40.4 ± 4.3 ***	50.3 ± 7.0 $^{***,\#}$

B.		freezing behavior during **selected calls**; analyzed with reduced number of frames	
rats with long 22-kHz calls and 44-kHz calls		freezing [%] during emission of:	
	no. of frames	long 22-kHz calls	44-kHz calls
with calls of ≥ 150 ms duration (n = 32)	3	49.5 ± 7.6	58.8 ± 8.0
with calls of ≥ 500 ms duration (n = 28)	5	67.3 ± 7.3	54.1 ± 9.1
	10	61.9 ± 8.1	53.3 ± 9.1
	15	52.6 ± 9.0	48.4 ± 9.4

Table 2.

Freezing associated with emission of long, monotonous vocalizations.

All Wistar rats which undergone 10 trials of fear conditioning were analyzed (**Tab. 1** /Exp. 1-3/#2,4,8,13; n = 46). **A.** Freezing (%) in 10-s-long bins where rats emitted exclusively long 22-kHz vocalizations vs. exclusively 44-kHz vocalizations. Results were compared to baseline freezing levels before conditioning (first 5 min) and during 10-s-long periods with no vocalizations (w/o calls). More information in the text. *** vs. “first 5 min”, $p < 0.001$; $^{\#}$ vs. “w/o calls”, $p < 0.05$; both Wilcoxon; NA, not analyzed. **B**. Freezing during the emission episodes of long 22-kHz and 44-kHz calls. Pairs of 44-kHz and long 22-kHz vocalizations were randomly selected from each animal. Freezing levels (%) did not differ between 22-kHz vs. 44-kHz calls (0.2054–0.7776 p levels, Wilcoxon). Minimum freezing duration used: 30 frames (**A**), 3 frames (for pairs of ≥ 150 ms vocalizations) or 5, 10, and 15 frames for ≥ 500 ms vocalizations (**B**).

typical for 22-kHz vocalizations. *Vice versa* we also observed 22-kHz calls with 44-kHz-like elements. Therefore, we propose that these long 22-kHz and 44-kHz vocalizations constitute a *supertype* group of long unmodulated aversive calls ("long 22/44-kHz vocalizations").

We observed a stable, approximately 1.5 ratio in peak frequency levels between 22-kHz and 44-kHz vocalizations within individual rats. Specifically, in fourteen rats (13 Wistar and 1 SHR) with a clear transition from 22-kHz to 44-kHz calls during the fear conditioning session (n = 14, selected from **Tab. 1**/Exp. 1-3/#2,4,6-8,10-13), the proportion between the frequencies of the long 22-kHz vocalizations and the long 44-kHz calls was 1.48 ± 0.02. Similar results were obtained for 70 step up (1.53 ± 0.03) and 65 step down (1.59 ± 0.02) 44-kHz calls – altogether suggesting a 1.5-times or 3:2 frequency ratio. This ratio and its relevance has been observed in invertebrates and vertebrates including human speech and music (Hoeschele, 2017). In music theory, 3:2 frequency ratio is referred to as a perfect fifth and is often featured, e.g., the first two notes of the *Star Wars* 1977 movie (ascending, i.e., step up; comp. **Fig. 2C**, Track 1) and *Game of Thrones* 2011 television series (descending, i.e., step down; comp. **Fig. 2D**, Track 2) theme songs. All of which may point to a common basis for this sound interval and its prevalence which could be explained by the observation that all physical objects capable of producing tonal sounds generate harmonic vibrations, the most prominent being the octave, perfect fifth, and major third (Christensen, 1993, discussed in Bowling and Purves, 2015).

New calls form separate, distinct group

Next, we showed that 44-kHz calls indeed constitute a distinct, separate type of ultrasonic vocalizations as it was sorted into isolated clusters by two different methods. First, using the DBSCAN algorithm method based on calls' peak frequency and duration, we were able to divide all vocalizations recorded during all training sessions into 44-kHz vocalizations vs. all other vocalizations as two separate clusters (**Fig. 3A**). Secondly, a clustering algorithm that includes call contours, i.e., k-means with UMAP projection done via DeepSqueak (**Figs 3BC**, **3S1**), sorted 44-kHz vocalizations of different subtypes including unusual ones (**Fig. 2S1A-F**), into topologically-separate groups. Notably, flat 44-kHz calls were consistently in a separate cluster from 22-kHz calls **Figs 3C**, **3S1B**).

Specific response to 44-kHz playback

To describe the behavioral and physiological impact of 44-kHz vocalizations, we performed playback experiments in two separate groups of rats (Methods, **Figs 4**, **4S1**). Overall, the responses to 44-kHz aversive calls presented from the speaker were either similar to 22-kHz vocalizations or in-between responses to 22-kHz and 50-kHz playbacks. For example, the heart rate of rats exposed to 22-kHz and 44-kHz vocalizations decreased, and increased to 50-kHz calls (**Fig. 4A**, comp. Olszyński et al., 2020). Whereas the number of vocalizations emitted by rats was highest during and after the playback of 50-kHz, intermediate to 44-kHz and lowest to 22-kHz playbacks (**Figs 4BC**, **4S1EF**). Additionally, the duration of 50-kHz vocalizations emitted in response to 44-kHz playback was also intermediate, i.e., longer than following 22-kHz playback (**Fig. 4D**) and shorter than following 50-kHz playback (**Figs 4D**, **4S1G**). Finally, similar tendencies were observed in the distance travelled and time spent in the half of the cage adjacent to the speaker (**Fig. 4S1A-D**).

Discussion

As Charles Darwin noted above (Darwin, 1872) and other researchers have confirmed (Briefer et al., 2012), the frequency level of animal calls is a vocal parameter that changes in accordance with its arousal state (intensity) or emotional valence (positive/negative state). The frequency shifts towards both higher and lower levels, i.e., alterations were observed during both positive

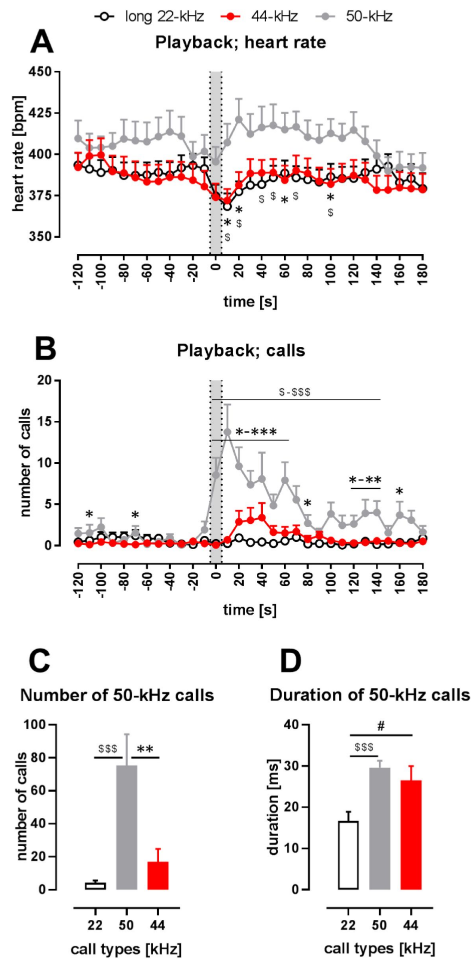

Fig. 4.

Physiological and behavioral response to playback of 44-kHz calls (vs. 50-kHz and 22-kHz calls) presented from a speaker to naïve Wistar rats.

A – heart rate (HR); **B** – the number of emitted vocalizations. **AB** – gray sections correspond to the 10-s-long ultrasonic playback. Each point is a mean for a 10-s-long time-interval with SEM. **CD** – properties of 50-kHz vocalizations emitted in response to ultrasonic playback, i.e., number of calls (**C**) and duration (**D**) calculated from the 0-120 s range. **A** – 50-kHz playback resulted in HR increase (playback time-interval vs. 10-30 s time-interval, p = 0.0007), while the presentation of the aversive playbacks resulted in HR decrease, both in case of 22-kHz (p < 0.0001) and 44-kHz (p = 0.0014, average from -30 to -10 time-intervals (i.e., "*before*") vs. playback interval, all Wilcoxon), which resulted in different HR values following different playbacks, especially at +10 s (p = 0.0097 for 50 kHz vs. 22-kHz playback; p = 0.0275 for 50 kHz vs. 44-kHz playback) and +20 s time-intervals (p = 0.0068, p = 0.0097, respectively, all Mann-Whitney). **B** – 50-kHz playback resulted also in a rise of evoked vocalizations (*before* vs. 10-30 s time-interval, p = 0.0002, Wilcoxon) as was the case with 44-kHz playback (p = 0.0176 in respective comparison), while no rise was observed following 22-kHz playback (p = 0.1777). However, since the increase in vocalization was robust in case of 50-kHz playback, the number of emitted vocalizations was higher than after 22-kHz playback (e.g., p < 0.0001 during 0-30 time-intervals) as well as after 44-kHz playback (e.g., p < 0.0001 during 0-10 time-intervals, both Mann-Whitney). Finally, when the increases in the number of emitted ultrasonic calls in comparison with *before* intervals were analyzed, there was a difference following 44-kHz vs. 22-kHz playbacks during 30 s and 40 s time intervals (p = 0.0420 and 0.0430, respectively, Wilcoxon). **C** – During the 2 min following the onset of the playbacks, rats emitted more ultrasonic calls during and after 50-kHz playback in comparison with 22-kHz (p < 0.0001) and 44-kHz (p = 0.0011) playbacks. The difference between the effects of 22-kHz and 44-kHz playbacks was not significant (p = 0.2725, comp. **Fig. 4S1F**; all Mann-Whitney). **D** – Ultrasonic 50-kHz calls emitted in response to playback differed in their duration, i.e., they were longer to 50-kHz (p = 0.0004) and 44-kHz (p = 0.0273, both Mann-Whitney) playbacks than to 22-kHz playback. * 50-kHz vs. 44-kHz, $ 50-kHz vs. 22-kHz, # 22-kHz vs. 44-kHz; one character (*, $ or #), p < 0.05; two, p < 0.01; three, p < 0.001; Mann-Whitney (**AB**) or Wilcoxon (**CD**). Values are means ± SEM, n = 13-16.

(appetitive) and negative (agonistic/aversive) situations, however, as a general rule, frequency usually increases with an increase in arousal (Briefer et al., 2012). We would like to propose a hypothesis that our prolonged fear conditioning increased the arousal of the rats with no change in the valence of the aversive stimuli.

It could also be argued that several factors, apart from increased arousal, contributed to the emergence of 44-kHz vocalizations in our fear-conditioned rats, e.g., heightened fear, stress/anxiety, annoyance/anger, disgust/boredom, grief/sadness, despair/helplessness, and weariness/fatigue. It is not possible, at this stage, to definitively determine which factors played a decisive role. Please note that the potential contribution of these factors is not mutually exclusive.

However, several arguments support the idea that 44-kHz vocalizations communicate an increased negative emotional state. First, in general, ultrasonic vocalizations serve as a means of communicating rats' emotional state (Brudzynski, 2013). Second, the changing of *the pitch of the voice bears some relation to certain states of feeling* (Darwin, 1872). Third, 44-kHz calls were notably more frequent during prolonged aversive stimulation, i.e., the 5th-10th trials of fear conditioning. Fourth, they were linked to freezing. Fifth, they appeared as partial replacements of, established as aversive, 22-kHz calls – in the presence of the same painful stimulus. Sixth, numerous instances of vocalizations featured both 22-kHz-like and 44-kHz-like call-elements.

Also, several observations contradict the potential contribution of fatigue. The sound mean power of 44-kHz vocalizations was comparable to, or possibly even higher than, that of 22-kHz calls, despite the higher energy costs associated with producing higher-pitched calls (Sonninen and Hurme, 1998), i.e., the rats emitting 44-kHz calls invested additional energy to communicate their emotional state; both *in vivo* measurements (Riede, 2013) and computer modelling (Hakansson et al., 2022) demonstrated that producing calls of higher frequency, such as 50 kHz vs. 22 kHz, requires increased activity of various muscles. Additionally, the mean power of 44-kHz vocalizations remained strong and stable for several trials – in contrast to 22-kHz vocalizations. Finally, when 44-kHz calls started to appear in significant numbers, i.e., after the 4th-5th trials of fear conditioning, they were as long as 22-kHz vocalizations.

Concerning the latter, we observed a significant decrease in the mean power of 22-kHz vocalizations during the fear conditioning session. Such reduction could potentially be attributed to fatigue (as observed in humans, Kitch and Oates, 1994), despair (e.g., as a reaction to the lack of effects from repeated emissions of 22-kHz calls), or both. The reduction in the amplitude of 22-kHz calls during the 10-trial fear conditioning was also recently observed by others (Gonzalez-Palomares et al., 2023).

Amounting research points to the utility of rat ultrasonic vocalizations to alter emotional states, evidenced by behavioral changes, in tested rats *via* playback of affectively valenced calls (Bonauto et al., 2023). We have exposed rats to 44-kHz playback along with 22-kHz and 50-kHz playback. The experimental design (see methods for details) allowed us to compare rats' responses to 22-kHz vs. 44-kHz playbacks especially – with 50-kHz playback used as a form of control or baseline. In general, the rats responded similarly to hearing 44-kHz calls as they did to hearing aversive 22-kHz calls, especially regarding heart-rate change, despite the 44-kHz calls occupying the frequency band of appetitive 50-kHz vocalizations. This is contrary to some observations (Saito et al., 2019) which suggested that frequency band plays the main role in rat ultrasound perception. Please factor in potential carry-over effects (resulting from hearing playbacks of the same valence in a row) in the differences between responses to 50-kHz vs. 22/44-kHz playbacks, especially, those observed before the signal (**Fig. 4AB**). Other responses to 44-kHz calls were intermediate, they fell between response levels to appetitive vs. aversive playback, which might signify some behavioral specificity and importance (or possibly confusion). These latter effects were similar in both playback experiments despite an array of methodological differences between them. Overall, these initial results raise further questions about how, ethologically, animals may interpret the variation in hearing 22-kHz vs. 44-kHz calls and integrate this interpretation in their responses.

The question also is, why have the 44-kHz vocalizations been overlooked until now? On one hand, long (or not that long as in Biały et al., 2019), frequency-stable high-pitch vocalizations have been reported before (e.g., Sales, 1979; Shimoju et al., 2020), notably as caused by intense cholinergic stimulation (Brudzynski and Bihari, 1990) or higher shock-dose fear conditioning (Wöhr et al., 2005). However, they have not been systematically defined, described, fully shown or demonstrated to be a separate type of vocalization. On the other hand, 44-kHz calls were likely omitted as the analyses were restricted to canonical groups, i.e. flat 22-kHz and short 50-kHz calls, with a sharp dividing frequency border between the two (e.g., Kalamari et al., 2021, Potasiewicz et al., 2020, Turner et al., 2019) or even a frequency 'safety gap' between 22-kHz and 50-kHz vocalizations (e.g., Silkstone and Brudzynski, 2019, Garcia et al., 2015). Moreover – many older bat-detectors had limited frequency-range detection (e.g., up to 40 kHz in Sales, 1991), when stress-evoked types of ultrasonic calls were being established. Finally, 44-kHz vocalizations are emitted much fewer than 22-kHz calls (**Fig. 1FG**).

Here we present introductory evidence that 44-kHz vocalizations are a separate and behaviorally-relevant group of rat ultrasonic calls. These results require further confirmations and additional experiments, also in form of replication, including research on female rat subjects. However, our results bring to awareness that rats employ these previously unrecognized, long, high-pitched and flat aversive calls in their vocal repertoire. Researchers investigating rat ultrasonic vocalizations should be aware of their potential presence and to not rely fully on automated detection of high vs. low-pitch calls.

Materials and Methods

Animals

Wistar rats (n = 167) were obtained from The Center for Experimental Medicine of the Medical University of Bialystok, Poland; spontaneously hypertensive rats (SHR, n = 80) and Sprague-Dawley rats (n = 16) were from Mossakowski Medical Research Institute, Polish Academy of Sciences, Poland. All rats were males, 7 weeks of age on arrival, randomly assigned into groups and cage pairs where appropriate; housed with a 12 h light-dark cycle, ambient temperature (22–25 °C) with standard chow and water provided *ad libitum*. The animals were left undisturbed for at least one week before any procedures, then handled at least four times for 2 min by each experimenter directly involved for one to two weeks. All procedures were approved by Local Ethical Committees for Animal Experimentation in Warsaw.

Animal details: groups of animals used

Trace fear conditioning experiment

Wistar rats, both single-housed (n = 14) and pair-housed (n = 20), were implanted with radiotelemetric transmitters for measuring heart rate in an ultrasonic vocalization playback experiment previously described by us (Olszyński et al., 2020) after which, at 13 weeks of age, half of them (n = 17) were fear-conditioned (10 shocks), while the other half (n = 17) served as controls (**Tab. 1**/Exp. 1/#1-4, n = 34).

Delay fear conditioning experiment, rats with transmitters

Wistar rats (n = 94) and SHR (n = 80) were implanted with a radiotelemetric transmitters one week before fear conditioning during which they received 0, 1, 6 or 10 shocks at 12 weeks of age (**Tab. 1**/Exp. 2/#5-12, n = 174). All the details are described in Olszyński et al. (2021) and Olszyński et al. (2022).

Delay fear conditioning experiment, rats without transmitters

Wistar rats were housed in pairs; were not implanted with radiotelemetric transmitters to eliminate the potential effect of surgical intervention on vocalization; they received 10 conditioning stimuli at 12 weeks of age (**Tab. 1**/Exp. 3/#13, n = 10) – same as in Olszyński et al. (2021) and Olszyński et al. (2022).

Playback experiment, rats with transmitters

Wistar rats (n = 29) were housed in pairs; all were implanted with a radiotelemetric transmitter one week before the playback experiment. At 12 weeks of age, one group (n = 13) heard 50-kHz appetitive vocalization playback while the other (n = 16) 22-kHz and 44-kHz aversive calls (for details see below).

Playback experiment, rats without transmitters

Sprague Dawley rats (n = 16) were housed in pairs, were not implanted with the transmitters, and received 22-kHz, 44-kHz, and 50-kHz ultrasonic vocalization playback at 8 weeks of age (see below).

Surgery, transmitter implantation, heart-rate registration

Radiotelemetric transmitters (HD-S10, Data Sciences International, St. Paul, MN, USA) were implanted into the abdominal aorta of rats in specified groups as previously described (Olszyński et al., 2020, Olszyński et al., 2021). An illustrative image with the surgery details can be found elsewhere (Figure 5 in Pestana-Oliveira et al., 2020; please note, tissue glue was used instead of cellulose patches and silk sutures). The signal was collected by receivers (RSC-1, Data Sciences International, St. Paul, MN, USA) as previously described (Olszyński et al., 2020, Olszyński et al., 2021, Olszyński et al., 2022). Readings were processed using Dataquest ART (version 4.36, Data Sciences International) for trace fear conditioning (**Tab. 1**/Exp. 1) and Ponemah (version 6.32, Data Sciences International) software for other experiments (**Tab. 1**/Exp. 2-3 and playback experiments).

Fear conditioning

All conditioning procedures were conducted in a chamber (VFC-008-LP, Med Associates, Fairfax, VT, USA) located in an outer cubicle (MED-VFC2-USB-R, Med Associates) equipped with an ultrasound CM16/CMPA condenser microphone (Avisoft Bioacoustics, Berlin, Germany). Ultrasonic vocalizations were recorded via Avisoft USGH Recorder (Avisoft Bioacoustics), and rat behavior was recorded via NIR monochrome camera (VID-CAM-MONO-6, Med Associates). All procedures were described in detail before (Olszyński et al., 2021, Olszyński et al., 2022).

Trace fear conditioning (**Tab. 1**/Exp. 1/#1-4, n = 34 rats) was performed similarly to some previous reports (e.g., Jahołkowski et al., 2009). Rats were individually placed in the fear conditioning apparatus in one of two different contexts: A (safe) or B (unsafe). Context A was in an illuminated room with the cage interior with white light, the cage floor was made of solid plastic, and the cage was scented with lemon odor, cleaned with a 10% ethanol solution; the experimenter was male wearing white gloves. Context B was a different, dark room, with the cage interior with green light, the floor was made of metal bars, and the cage was scented with mint odor, cleaned with 1% acetic acid; the experimenter was female with violet gloves. The procedure: on day -2, each rat was habituated to context A for 20 min; on day -1, habituated to context B for 20 min; on day 0, each rat was placed for 52 min in context A; on day 1, after 10 min in context B, the rat received 10 conditioning stimuli (15-s-long sine wave tone, 5 kHz, 85 dB) followed by a 30 s trace period and a foot-shock (1 s, 1 mA) and 210 s inter-trial interval, i.e., ITI; total session duration: 52 min. Control rats were subjected to the same procedures but did not receive the electric shock at the end of trace periods. The animals were tested with the same protocol without shocks in context A (day 2) and context B (day 3). During the test session, control animals showed a lower level of freezing than conditioned animals (1.3 ± 0.8% vs. 19.7 ± 4.3% during the first 5 min in unsafe context B and 0.4 ± 0.3% vs. 9.9 ± 1.9% during 10 s following the time of expected shock in context B, results averaged from the first 3 out of 10 trials; p = 0.0003 and p = 0.0001, respectively, Mann-Whitney); none of the control animals emitted 44-kHz calls, neither the fear conditioning day nor the test days.

Delay fear conditioning. (**Tab. 1**/Exp. 2-3/#5-13, n = 184 rats) The procedure and its results were described before (Olszyński et al., 2021, Olszyński et al., 2022); rats received 1, 6 or 10 conditioning stimuli (20-s-long white light co-terminating with an electric foot-shock, 1 s, 1 mA). For control rats, an equal time-length procedure was done for each conditioning protocol, i.e., the same parameters as in 1, 6 or 10 stimuli groups, with no shock. Control animals showed a lower level of freezing than conditioned animals. There were only 4 ultrasonic calls we classified as 44-kHz vocalizations among 4,126 vocalizations emitted by the control rats during training and testing. We did not observe any difference in the number of 44-kHz vocalizations between Wistar rats with transmitters vs. without transmitters during delay conditioning training (p = 0.8642, Mann-Whitney). These two groups were therefore reported together.

Measuring freezing

Freezing behavior was scored automatically using Video Freeze software (Med Associates) with a default motion index threshold of 18. To avoid including brief moments of the animal's stillness, freezing was measured only if the animal did not move for at least 1 s, i.e., 30 video frames, with some exceptions, see next.

Vocalization-nested freezing behavior

Freezing at the exact times of ultrasonic calling was measured in rats that had undergone 10 trials of fear conditioning which produced 44-kHz calls (n = 32, selected from **Tab. 1**/Exp. 1-3/#2,4,8,13). From each rat, one 44-kHz call was randomly selected along with the long 22-kHz call closest to it. Such pairs of vocalizations were selected with either ≥150 ms duration (n = 32) or ≥500 ms duration (n = 28). For each pair of vocalizations, the freezing behavior was calculated from the entire duration of the shorter call and for the equal-time-length period in the middle of the longer vocalization. Due to the shortened time-scale, the minimal freezing detection window was reduced to 3 frames for ≥150-ms-long calls as well as 5, 10, and 15 frames – for ≥500-ms calls.

Ultrasonic playback

It was performed as described previously (Olszyński et al., 2020, Olszyński et al., 2021, Olszyński et al., 2022) in individual experimental cages with acoustic stimuli presented through a Vifa ultrasonic speaker (Avisoft Bioacoustics, Berlin, Germany) connected to an UltraSoundGate Player 116 (Avisoft Bioacoustics). Ultrasonic vocalizations emitted by the rat were recorded by a CM16/CMPA condenser microphone (Avisoft Bioacoustics). Both playback and recording of calls were performed using Avisoft Recorder USGH software (version 4.2.28, Avisoft Bioacoustics). The locomotor activity was recorded with an acA1300-60gc camera (Basler AG, Ahrensburg, Germany). There were 8 sets of ultrasonic vocalizations presented:

1. **44-kHz long calls**, 8 calls in 1 repeat, constant frequency (2.7 ± 0.1 kHz max-min frequency difference), 42.1 ± 0.2 kHz peak frequency, 1064.3 ± 89.6 ms duration with 199.0 ± 14.7 ms sound intervals;
2. **22-kHz long calls**, 8 calls in 1 repeat, typical long 22-kHz vocalizations, constant frequency (1.9 ± 0.9 kHz max-min frequency difference), 24.5 ± 0.2 kHz peak frequency, 1066.4 ± 90.2 ms duration with 195.6 ± 15.5 ms sound intervals;
3. **22-kHz short modulated calls**, 26 calls in 2 repeats, short (<300 ms), not resembling typical 22-kHz long calls (5.3 ± 0.4 kHz max-min frequency difference), 22.7 ± 0.6 kHz peak frequency, 24.7 ± 1.6 ms duration with 172.8 ± 5.6 ms sound intervals;
4. **22-kHz short flat calls**, 43 calls in 1 repeat, short (<300 ms), resembling typical 22-kHz long calls, constant frequency (2.3 ± 0.1 kHz max-min frequency difference), 25.1 ± 0.3 kHz peak frequency, 102.4 ± 10.9 ms duration with 132.1 ± 6.2 ms sound intervals;
5. **50-kHz modulated calls**, 23 calls in 2 repeats, moderately modulated (8.6 ± 0.3 kHz max-min frequency difference), 61.0 ± 0.8 kHz peak frequency, 37.6 ± 1.5 ms duration with 183.7 ± 4.5 ms sound intervals;
6. **50-kHz flat calls**, 29 calls in 2 repeats, constant frequency (4.2 ± 0.2 kHz max-min frequency difference), 53.5 ± 0.5 kHz peak frequency, 66.2 ± 3.8 ms duration with 144.1 ± 4.4 ms sound intervals;
7. **50-kHz trill calls**, 29 calls in 2 repeats, highly modulated (37.4 ± 1.7 kHz max-min frequency difference), 68.0 ± 0.9 kHz peak frequency, 53.7 ± 1.4 ms duration with 158.5 ± 4.9 ms sound intervals;
8. **50-kHz kHz mixed calls**, used previously in Olszyński et al. (2020), Olszyński et al. (2021), and Olszyński et al. (2022), 28 calls, in 3 repeats, frequency modulated and trill subtypes, 9.8 ± 1.9 kHz max-min frequency difference, 58.6 ± 0.7 kHz peak frequency, 28.4 ± 1.6 ms duration with 91.4 ± 1.4 ms sound intervals.

Calls were presented with a sampling rate of 250 kHz in 16-bit format. All calls except for 50-kHz mixed calls were collected in our laboratory from fear conditioning or playback experiments. Calls in the same set were taken from one animal wherever possible. The sound interval was adjusted if it was peculiarly long or the sequence was interrupted by other types of calls in the original recordings.

Playback procedure, rats with transmitters; as previously described (Olszyński et al., 2020, Olszyński et al., 2021, Olszyński et al., 2022). Before playback presentation, animals were habituated for 3 min to the experimental conditions, i.e., recording cage, presence of the speaker and microphone, over 4 days. Habituated rats then underwent a playback procedure, in short, after 10 min of silence, the rats were exposed to four 10-s-long call sets (either aversive or appetitive) with 5-min-long ITI in-between; a rat that received appetitive playback was followed by a rat receiving aversive playbacks etc. Also, the order of the presented sets was randomized between animals. The aversive-calls playback contained sets nos. 1-4. The appetitive-calls playback contained sets nos. 5-8. Since initial analysis showed no differences within responses to 22-kHz aversive sets and within responses to 50-kHz appetitive sets, we decided to show the results following playback of 44-kHz long calls (set no. 1), 22-kHz long calls (set no. 2), and 50-kHz modulated calls (set no. 5) only.

Playback procedure, rats without transmitters. Before playback presentation, animals were habituated for 3 min to the experimental conditions, i.e., recording cage, presence of the speaker and microphone, over 4 days. After 5 min of initial silence, the rats were presented with two 10-s-long playback sets of either 22-kHz (set no. 2; n = 8) or 44-kHz calls (set no. 1; n = 8), followed by one 50-kHz modulated call 10-s set (no. 5) and another two playback sets of either 44-kHz or 22-kHz calls not previously heard. The playback presentations were separated by 3 min ITI. Responses to the pairs of playback sets were averaged.

Locomotor activity in playback. An automated video tracking system (Ethovision XT 10, Noldus, Wageningen, The Netherlands) was used to measure the total distance travelled (cm). Proximity to the speaker was expressed as the percentage of time spent in the half of the cage closer to the ultrasonic speaker. Center-point of each animal's shape was used as a reference point for measurements of locomotor activity thus registering only full-body movements.

Analysis of ultrasonic vocalizations

Audio recordings were analyzed manually using SASLab Pro (version 5.2.xx, Avisoft Bioacoustics) as described (Olszyński et al., 2020, Olszyński et al., 2021, Olszyński et al., 2022) to measure key features of calls and categorize them into subtypes.

Sound mean power was measured as the average spectra power density of the vocalization contour using DeepSqueak software. Initially, calls were detected using the default rat long-vocalization neural network (Long Rat Detector YOLO R1) and subsequently manually reviewed and corrected where necessary. We analyzed a subset of Wistar rats subjected to 10-trial fear conditioning that emitted more than 20 instances of 44-kHz calls during the fear conditioning session (n = 17, selected from **Tab. 1**/Exp. 1-3/#2,4,8,13). It is important to note that due to the directional characteristics of the microphones used, angular attenuation occurred during audio recording. This phenomenon results in a selective reduction in the intensity of higher frequency sounds, dependent on the angle between the sound emitter and the microphone (as specified in the CM16/CMPA microphone hardware specification page, Avisoft Bioacoustics website). In our experimental setup, we approximated a 45° angle between the plane of the rat's head and the plane of the microphone's membrane. This angle corresponds to an estimated 10 dB attenuation (adopting a conservative estimate) of 40-kHz frequencies compared to 20-kHz frequencies for which there is even a small dB gain due to these hardware properties, 44-kHz calls are predicted to be approximately at least 10 dB louder in reality than what was recorded.

22-kHz vs. 44-kHz frequency ratio. A clear transition point between 22-kHz and 44-kHz long calls was observed in n = 13 Wistar rats and n = 1 SHR. In each case, ten 22-kHz calls followed by ten 44-kHz calls were analyzed (n = 14, selected from **Tab. 1**/Exp. 1-3/#2,4,6-8,10-13).

Step up and step down frequency ratio. Rats which emitted at least five vocalizations of the specific subtype were analyzed (step up, n = 14; step down, n = 13; selected from **Tab. 1**/Exp. 1-3/#2,4,7,8,13; 5 calls of the two subtypes from each rat were chosen randomly and the frequencies of their elements were measured.

Ultrasonic vocalizations clustering (two independent methods)

Calls of conditioned and control animals were taken from all fear conditioning training sessions (**Tab. 1**/Exp. 1-3, n = 218). We used **DBSCAN algorithm** (Ester et al., 1996); a density based method, from the scikit-learn (sklearn) Python package, because of its ability to detect a desired number of clusters of arbitrary shape; with two main input parameters: MinPts (minimal number of points forming the core of the cluster) and ε (the maximum distance two points can be from one another while still belonging to the same cluster). To avoid detecting small clusters, we limited MinPts to 150 samples. The heuristic method described by Ester et al. (Ester et al., 1996) was implemented to find the initial range of ε. All the input data were standardized. The silhouette coefficient (Rousseeuw, 1987) was used to control the quality of the clustering. Maximizing ε among different ranges helped to select the most relevant number of identified clusters. Clustering with ε in the range of 0.14–0.2 resulted in a silhouette coefficient around 0.2–0.5.

K-means algorithm. Vocalizations of selected fear-conditioned rats with 6-10 shocks and >30 of 44-kHz calls (n = 26, selected from **Tab. 1**/Exp. 1-3/#2,4,7,8,11-13) were detected using a built-in neural network for long rat calls (Long Rat Detector YOLO R1) on DeepSqueak (Coffey et al., 2019) software (version 3.0.4) running under MATLAB (version 2021b, MathWorks, Natick, MA, USA) and manually revised for missed and mismatched calls. Unsupervised k-means clustering was based on call contour, frequency and duration variables, with equal weights assigned, and several descending elbow optimization parameters were used to obtain different maximum numbers of clusters together with Uniform Manifold Approximation and Projection for Dimension Reduction (UMAP) (McInnes et al., 2018) for superimposing and visualization of clusters.

Quantification and statistical analysis

Data were analyzed using non-parametric Friedman, Wilcoxon, Mann-Whitney tests with GraphPad Prism 8.4.3 (GraphPad Software, San Diego, CA, USA); the p values are given, $p < 0.05$ as the minimal level of significance. Figures were prepared using the same software and depict average values with a standard error of the mean (SEM).

Data availability

Raw data (calls' peak frequency and duration) analyzed, ultrasonic playback files used (.wav), data supporting clustering files for DBSCAN (.csv), and extracted call contours for k-means (.mat) have been deposited to Mendeley Data at http:/to be provided/. The other data in this study are available from the corresponding author upon request.

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Author contributions

K.H.O, and R.P., and R.K.F. designed the study and wrote the manuscript. K.H.O., R.P., A.D.W., A.W.G., and O.G. performed the experiments. W.P. and M.K. performed DBSCAN analysis. R.P. performed k-means analysis. K.H.O., R.P., I.A.Ł., and A.D.W. analyzed the data. R.K.F. acquired the funding and supervised the project. All authors reviewed and approved the final version of the manuscript.

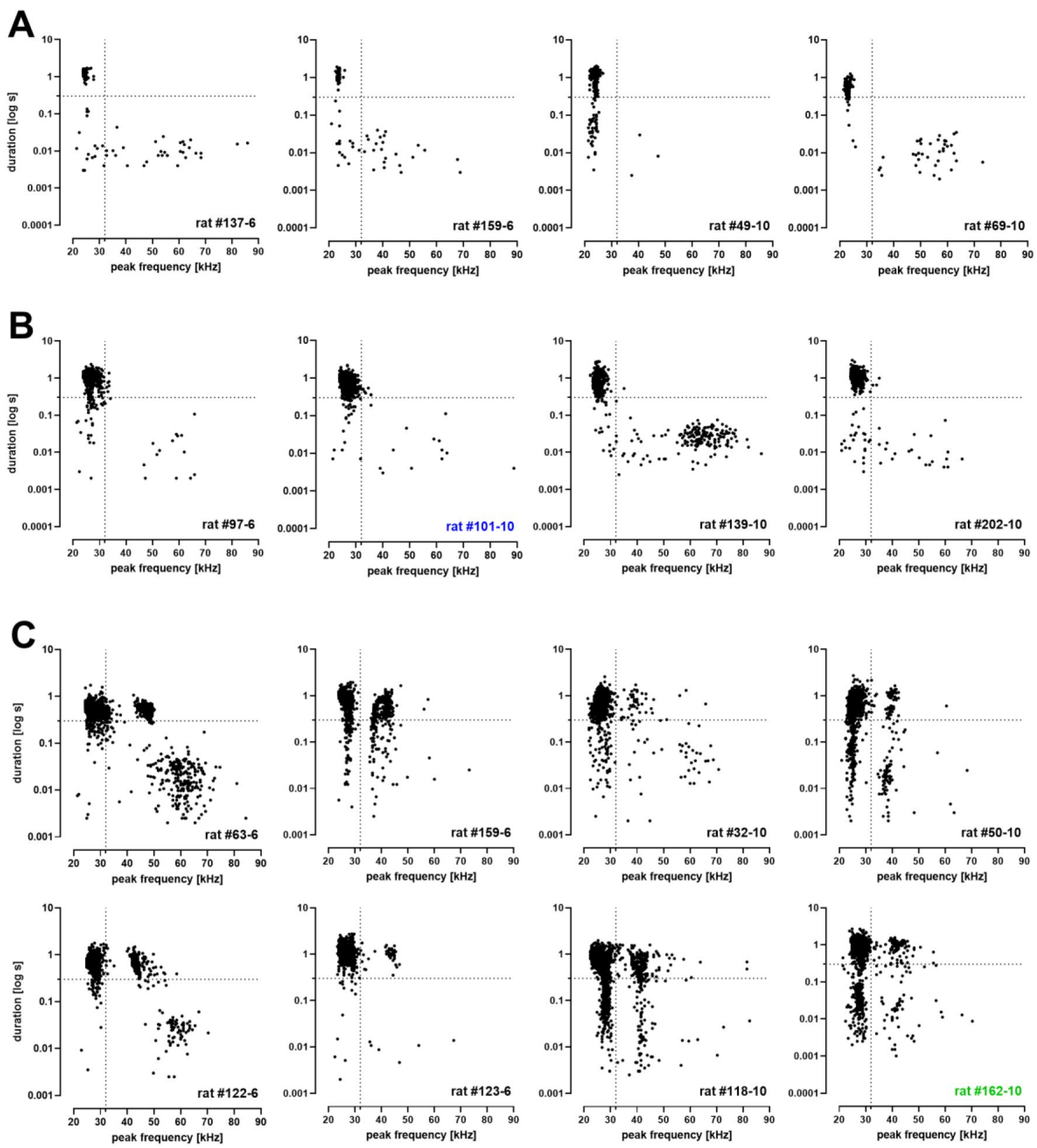

Fig. 1S1.

Variations of call frequency; shown in relation to call duration in Wistar rats that undergone 6 or 10 trials of delay fear conditioning (n = 16, selected from Tab. 1 /Exp. 2-3/#7,8,13).

Vocalizations plotted in relation to peak frequency (x axis) and duration (y axis). Each point corresponds to one vocalization. Vertical dotted line marks threshold value (32 kHz) between 22-kHz and 50-kHz calls. Horizontal dotted line marks threshold value (300 ms) between short and long 22-kHz calls (Brudzynski et al., 1993). Rat identifier is given in lower right corner; the number after dash indicates the number of conditioning trials. **A** – examples from four rats which emitted typical long 22-kHz calls (no 44 kHz calls). **B** – four typical long 22-kHz vocalizations with few long 22-kHz calls crossing the 32 kHz threshold. **C** – eight sample rats which emitted typical long 22-kHz vocalizations and atypical high-frequency aversive calls forming a distinct 44-kHz group.

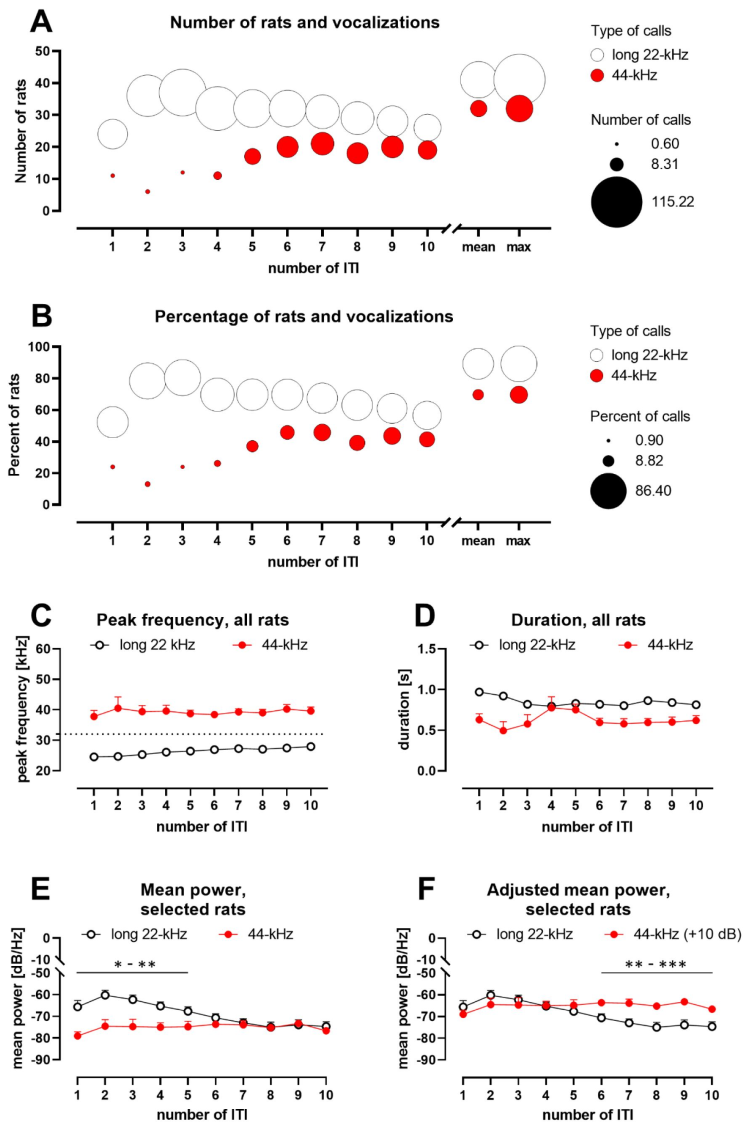

Fig. 1S2.

Changes in distribution (AB), frequency (C), duration (D), and mean power (EF) of long aversive vocalizations throughout fear conditioning session.

Data were acquired from all Wistar rats subjected to a 10-trial fear conditioning procedure (**Tab. 1**/Exp. 1-3/#2,4,8,13; n = 46). X-axes represent subsequent inter-trial intervals (ITI) numbered after the preceding conditioned stimulus. **AB**. **Number or percentage of rats emitting long vocalizations.** Bubbles represent long 22-kHz calls (white) or 44-kHz calls (red); bubble size scales with the amount of vocalizations. Emission of 44-kHz calls and the number of animals emitting them increases in the latter half of the session. Data are absolute values (**A**) or percentages (**B**); "mean", average value from all ITI, "max", maximum values from each rat. **C**. **Frequencies of 22-kHz and 44-kHz vocalizations.** Horizontal dotted line marks the threshold value (32 kHz) between 22-kHz and 50-kHz/44-kHz calls. Peak frequency of long vocalizations rose gradually in all rats. **D. Duration of 22-kHz and 44-kHz vocalizations.** The duration of 22-kHz calls gradually declined. The duration of 44-kHz calls peaked after the 4th ITI. **E. Mean power of 22-kHz and 44-kHz vocalizations.** Mean power spectral density (loudness, amplitude) of 22-kHz calls, n = 14-17 per ITI; and 44-kHz calls, n = 5-17 per ITI; **F.** results for 44-kHz vocalizations (from **E**) were adjusted for angular attenuation, i.e., +10 dB. Before the adjustment: during the first half of the session, 22-kHz calls appeared louder than 44-kHz calls, in the second half of the session the difference dissipated. After the adjustment: both types of calls started on a comparable amplitude level, but in the 6th-10th ITI, 22-kHz calls became quieter than 44-kHz calls. Values are means ± SEM (**C-F**). Graphs show either all rats (**A-D**, n = 46) or rats which met the criteria of emitting >20 of 44-kHz calls (**EF**, n = 17 selected from n = 46); *p < 0.05, **p < 0.01, ***p < 0.001).

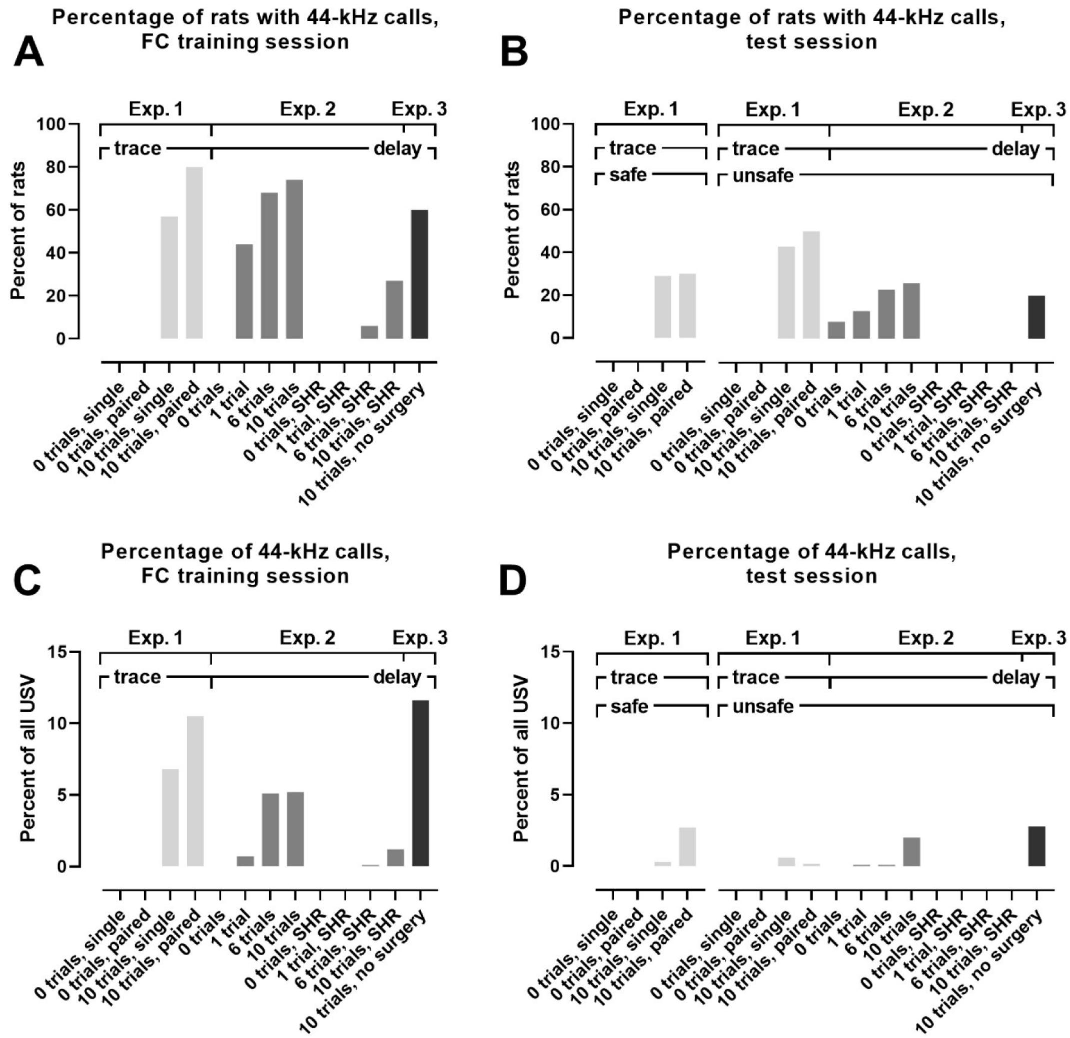

Fig. 1S3.

Percentage of animals emitting 44-kHz calls (AB) and percentage of 44-kHz calls in all vocalizations (CD) emitted by Wistar rats and SHR.

Results from three main fear conditioning experiments are shown (comp. **Tab. 1**), i.e., Exp. 1 (light gray bars), Exp. 2 (dark gray bars), and Exp. 3 (black bars), which all were performed with Wistar rats or SHR (when specified in the x-axis labels). The labels denote different experimental groups used across the experiments (see **Tab. 1** for the number of animals in each group). Results were obtained during fear conditioning training (**A**, **C**) and testing sessions (**B**, **D**). Rats subjected to trace fear conditioning were tested in safe and unsafe contexts, while in delay fear conditioning, the rats were tested only in an unsafe context (see Methods). 44-kHz calls appeared most often in Wistar rats which had undergone 10-trial fear conditioning procedures. Please note that the experiments were not performed in parallel.

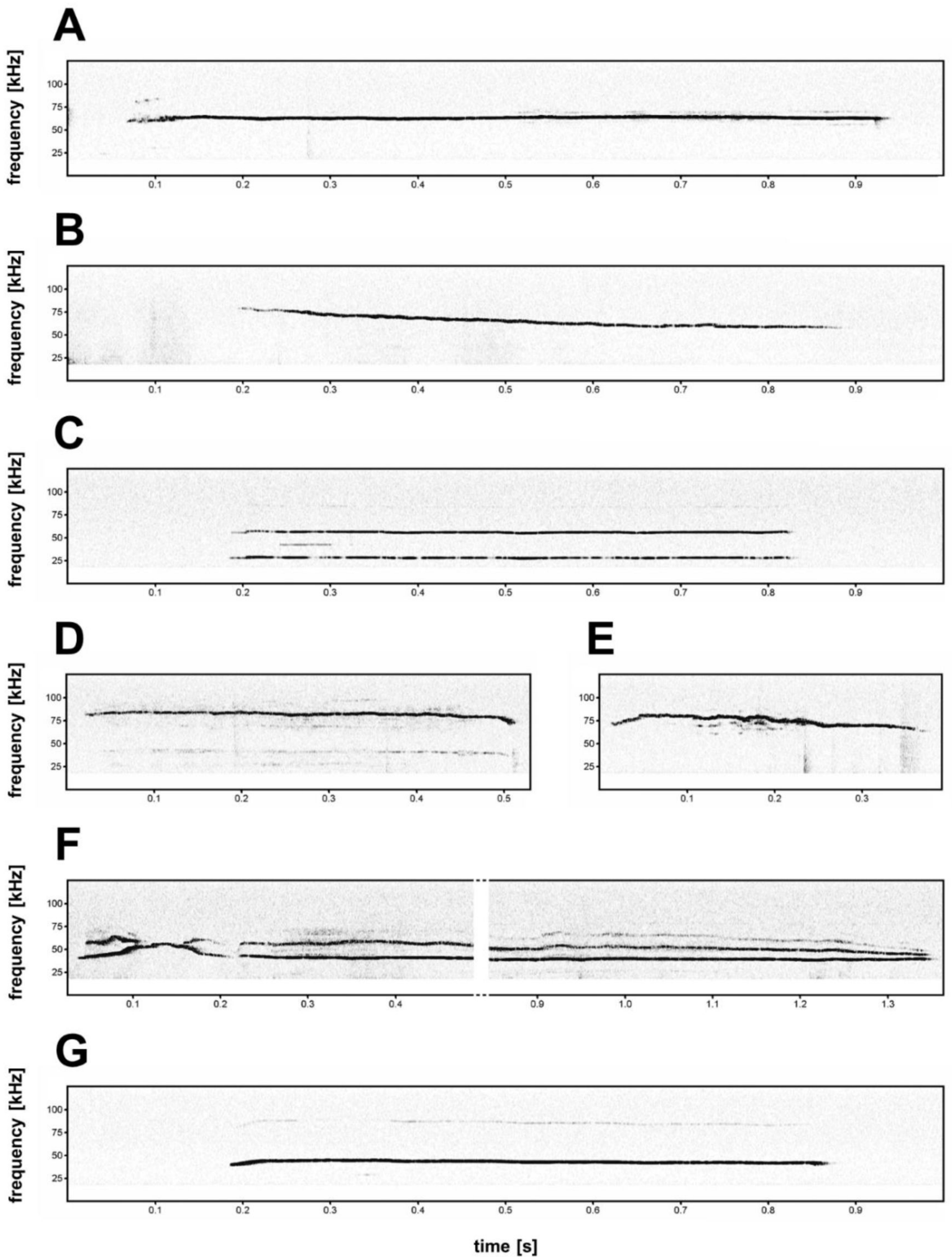

Fig. 2S1.

Non-typical 44-kHz aversive vocalizations.

A, **B** – constant frequency calls with very high peak frequency (**A**, peak frequency = 62.9 kHz; **B**, peak frequency = 65.9 kHz, start peak frequency = 78.1 kHz). **C**, **D** – harmonic aversive vocalizations, where element with fundamental frequency (F0, lowest frequency of the vocalization) is not with maximum amplitude, i.e., peak frequency is determined from the higher call component (**C**, F0 = 27.8 kHz, peak frequency = 55.6 kHz; **D**, F0 = 40 kHz, peak frequency = 81.5 kHz). **E**, **F** – vocalizations with prominent duration but with modulated frequency (**E**, peak frequency = 69.3 kHz; **F**, peak frequency = 39.0 kHz). **A**, **G** – constant frequency calls from SHR (**G**, flat 44-kHz call, peak frequency = 42.4 kHz).

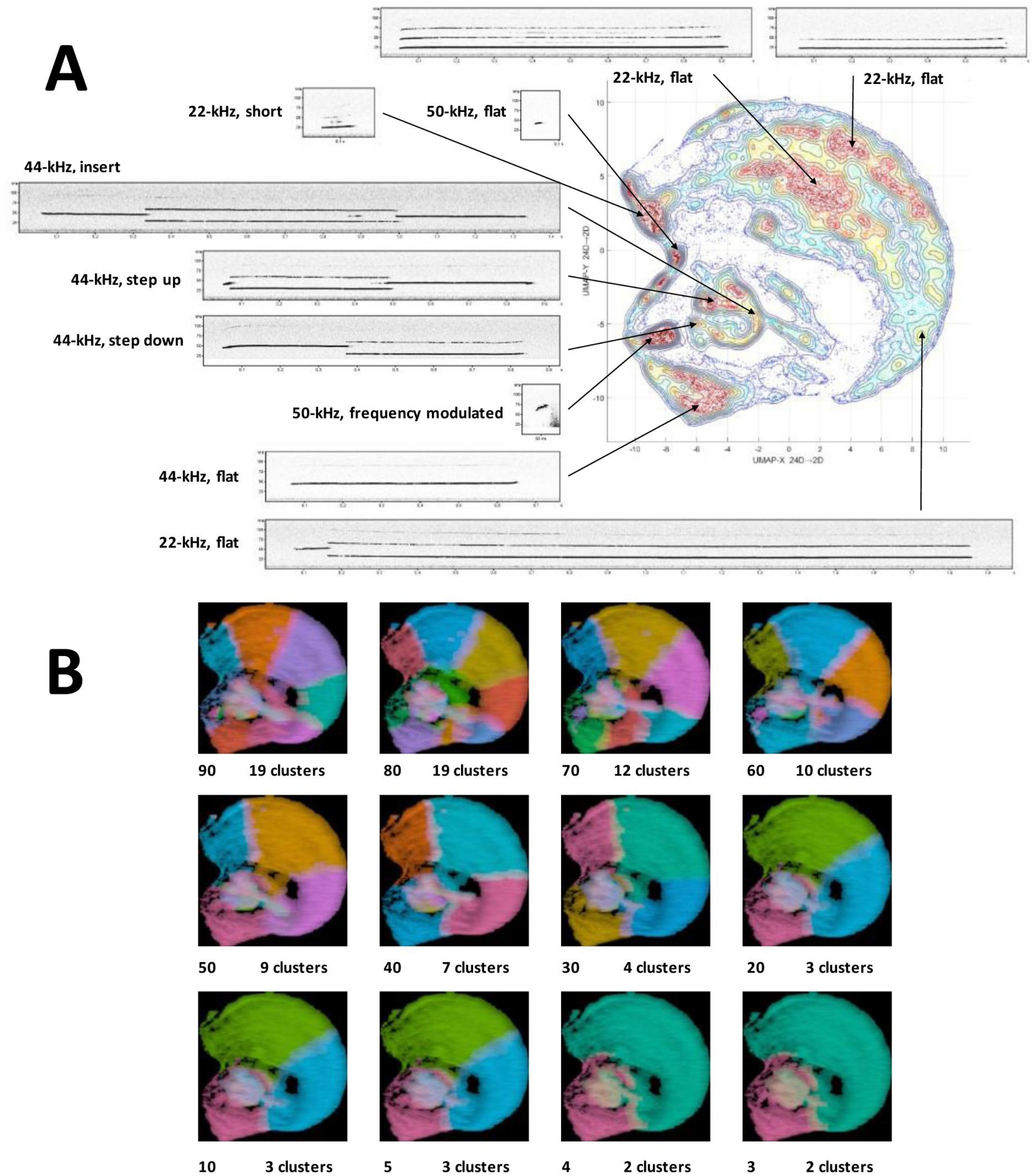

Fig. 3S1.

Clustering of ultrasonic vocalizations from rats emitting 44-kHz calls using UMAP projection and k-means.

A – topological plot of ultrasonic calls using UMAP embedding from selected rats emitting 44-kHz vocalizations during trace and delay fear conditioning training (n = 26, selected from **Tab. 1** /Exp. 1-3/#2,4,7,8,11-13), total number of calls n = 40,084, with spectrogram miniatures pointing to the general location from which they originated. **B** – comparison of unsupervised k-means clustering with different maximum possible number of clusters using elbow optimization (different clusters denoted by colors) done by DeepSqueak software, superposed over UMAP topological plot, number on the bottom left of the miniature denotes the maximum possible number of clusters set for elbow optimization, number on the bottom right denotes the resulting number of clusters after elbow optimization.

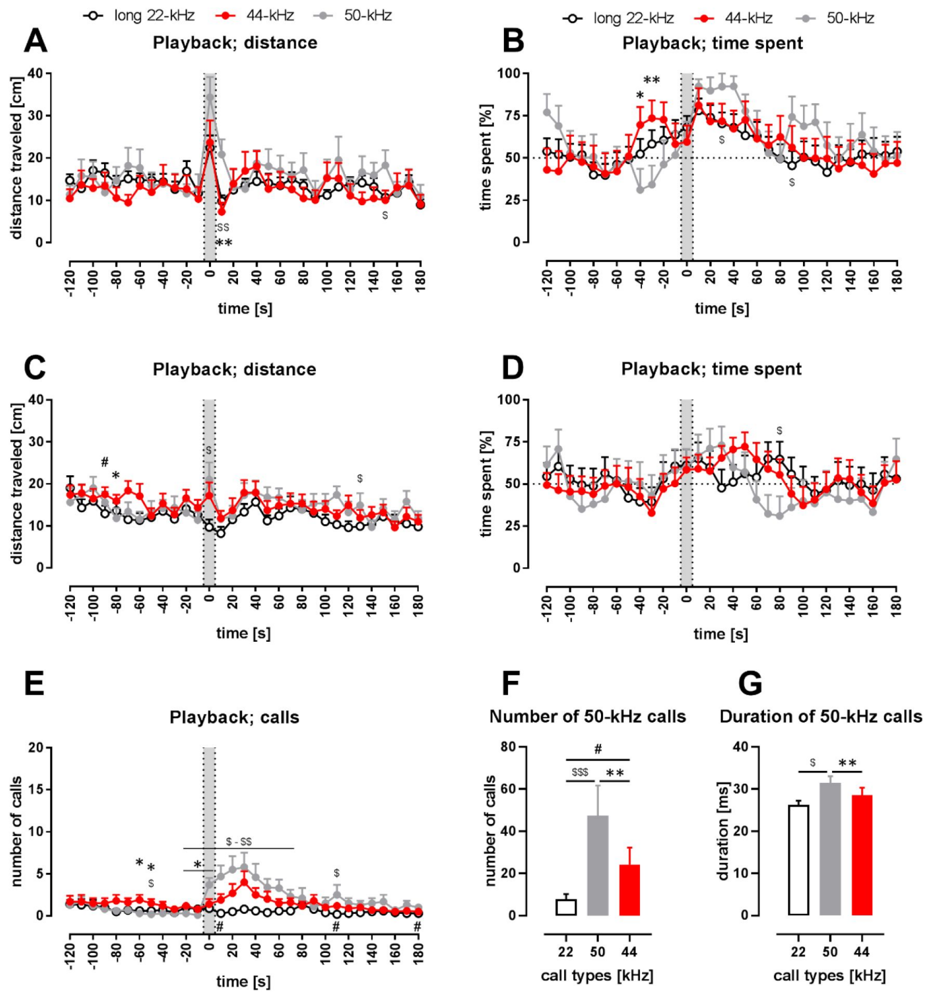

Fig. 4S1.

Behavioral response to playback of 44-kHz calls (vs. 50-kHz and 22-kHz calls).

AB – rats with implanted heart-rate transmitters (comp. **Fig. 4**), Wistar, n = 13-16; **C-G** – rats without transmitters, Sprague-Dawley, n = 15; **AC** – distance traveled; **BD** – time spent in the speaker's half of the cage; the dotted horizontal line marks a 50% chance value for time in a side of the cage; **E** – number of emitted vocalizations; **A-E** – gray sections correspond to the 10-s-long ultrasonic presentation, each point is a mean for a 10-s-long time-interval with SEM. **FG** – properties of 50-kHz vocalizations emitted in response to ultrasonic playback, i.e., number of calls (**F**) and duration (**G**) in 0-120 s range. **A-D** – playback presentation resulted in increased motor activity in case of, especially, 50-kHz playback and 44-kHz playback. Also, all kinds of playback resulted in increased time spent in the half of the cage next to the speaker. **E** – 50-kHz playback resulted in a rise of the number of evoked vocalizations (average from -30 to -10 time-intervals aka *before* vs. 10-30 s time-interval, p = 0.0010) as was the case with 44-kHz playback (p = 0.0142), respectively, while no rise was observed following 22-kHz playback (p = 0.2271, all Wilcoxon). However, since the increase in vocalization was robust in case of 50-kHz playback, the number of emitted vocalizations was higher than both after 22-kHz playback (e.g., p < 0.01 during 0-20 time-intervals) and after 44-kHz playback (p = 0.0172, 0 s time-interval, all Mann-Whitney). Finally, when the increases in the number of emitted ultrasonic calls in comparison with *before* intervals were analyzed, there was a difference following 44-kHz vs. 22-kHz playbacks during the 40 s time interval (p = 0.0017, Wilcoxon, comp. **Fig. 4B**). **F** – During the 2 min following the onset of the playbacks, the rats emitted more ultrasonic calls during and after 50-kHz playback in comparison with 22-kHz (p = 0.0002) and 44-kHz (p = 0.0067) playbacks; also, the rats emitted more ultrasonic calls during and after 44-kHz playback in comparison with 22-kHz playback (p = 0.0369), comp. **Fig. 4C**; all Wilcoxon). **G** – Ultrasonic 50-kHz calls emitted in response differed also in their duration, i.e., they were shorter to 22-kHz (p = 0.0195) and 44-kHz (p = 0.0039) playbacks than to 50-kHz playback. The difference between the effects of 22-kHz and 44-kHz playbacks was not significant (p = 0.5469, comp. **Fig. 4D**; all Wilcoxon). * 50-kHz vs. 44-kHz, $ 50-kHz vs. 22-kHz, # 22-kHz vs. 44-kHz; one character (*, $ or #), p < 0.05; two, p < 0.01; three, p < 0.001; Mann-Whitney (**AB**) or Wilcoxon (**CD**). Values are means ± SEM.

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Reviewer #1 (Public Review):

Olszyński and colleagues present data showing variability from canonical "aversive calls", typically described as long 22 kHz calls rodents emit in aversive situations. Similarly long but higher-frequency (44 kHz) calls are presented as a distinct call type, including analyses both of their acoustic properties and animals' responses to hearing playback of these calls. While this work adds an intriguing and important reminder, namely that animal behavior is often more variable and complex than perhaps we would like it to be, there is some caution warranted in the interpretation of these data.

The exclusive use of males is a major concern lacking adequate justification and should be disclosed in the title and abstract to ensure readers are aware of this limitation. With several reported sex differences in rat vocal behaviors this means caution should be exercised when generalizing from these findings. The occurrence of an estrus cycle in typical female rats is not justification for their exclusion. Note also that male rodents experience great variability in hormonal states as well, distinguishing between individuals and within individuals across time. The study of endocrinological influences on behavior can be separated from the study of said behavior itself, across all sexes. Similarly, concerns about needing to increase the number of animals when including all sexes are usually unwarranted (see Shansky [2019] and Phillips et al. [2023]).

Regarding the analysis where calls were sorted using DBSCAN based on peak frequency and duration, my comment on the originally reviewed version stands. It seems that the calls are sorted by an (unbiased) algorithm into categories based on their frequency and duration, and because 44kHz calls differ by definition on frequency and duration the fact that the algorithm sorts them as a distinct category is not evidence that they are "new calls [that] form a separate, distinct group". I appreciate that the authors have softened their language regarding the novelty and distinctness of these calls, but the manuscript contains several instances where claims of novelty and specificity (e.g. the subtitle on line 193) is emphasized beyond what the data justifies.

The behavioral response to call playback is intriguing, although again more in line with the hypothesis that these are not a distinct type of call but merely represent expected variation in vocalization parameters. Across the board animals respond rather similarly to hearing 22 kHz calls as they do to hearing 44 kHz calls, with occasional shifts of 44 kHz call responses to an intermediate between appetitive and aversive calls. This does raise interesting questions about how, ethologically, animals may interpret such variation and integrate this interpretation in their responses. However, the categorical approach employed here does not address these questions fully.

I appreciate the amendment in discussing the idea of arousal being the key determinant for the increased emission of 44kHz, and the addition of other factors. Some of the items in this list, such as annoyance/anger and disgust/boredom, don't really seem to fit the data. I'm not sure I find the idea that rats become annoyed or disgusted during fear conditioning to be a particularly compelling argument. As such the list appears to be a collection of emotion-related words, with unclear potential associations with the 44kHz calls.

Later in the Discussion the authors argue that the 44kHz aversive calls signal an increased intensity of a negative valence emotional state. It is not clear how the presented arguments actually support this. For example, what does the elongation of fear conditioning to 10 trials have to do with increased negative emotionality? Is there data supporting this relationship between duration and emotion, outside anthropomorphism? Each of the 6 arguments presented seems quite distant from being able to support this conclusion.

In sum, rather than describing the 44kHz long calls as a new call type, it may be more accurate to say that sometimes aversive calls can occur at frequencies above 22 kHz. Individual and situational variability in vocalization parameters seems to be expected, much more so than all members of a species strictly adhering to extremely non-variable behavioral outputs.

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Author response:

The following is the authors' response to the original reviews.

We would like to express our gratitude to the reviewers for their suggestions and critiques as we continually strive to enhance the quality of the manuscript. We improved it, by incorporating the reviewers' suggestions, changing the content and numbering of figures (Figs 1, 3S1 were edited; 4 figures were moved to supplemental materials), and adding several analyses suggested by the reviewers along with accompanying figures (1S2, 1S3) and tables (1 and 2). These analyses include investigating the link between freezing behavior and 44-kHz calls as well as their sound mean power and duration. Also, we have introduced detailed information regarding the experiments performed as well as expanded the description and discussion of the results section. Finally, we added the information about 44-kHz calls reported by another group – which was inspired by our findings.

Below is the point-by-point response to the reviewers' comments.

> ***Reviewer #1 (Public Review):***
>
> *Olszyński and colleagues present data showing variability from canonical "aversive calls", typically described as long 22 kHz calls rodents emit in aversive situations. Similarly long but higher-frequency (44 kHz) calls are presented as a distinct call type, including analyses both of their acoustic properties and animals' responses to hearing playback of these calls. While this work adds an intriguing and important reminder, namely that animal behavior is often more variable and complex than perhaps we would like it to be, there is some caution warranted in the interpretation of these data. The authors also do not provide adequate justification for the use of solely male rodents. With several reported sex differences in rat vocal behaviors this means caution should be exercised when generalizing from these findings.*

We fully agree that our data should be interpreted with caution and we followed the Reviewer's suggestions along these lines (see below). Also, we appreciate the suggestion to explore the prevalence of 44-kHz calls in female subjects, which would indeed represent an important and intriguing extension of our research. However, due to present financial constraints, we can only plan such experiments. To address the comment, we have added the sentence: "Here we are showing introductory evidence that 44-kHz vocalizations are a separate and behaviorally-relevant group of rat ultrasonic calls. These results require further confirmations and additional experiments, also in form of repetition, including research on female rat subjects."

It is important to note that the data presented in the current manuscript originates primarily from previously conducted experiments. These earlier experiments employed male subjects only; it was due to established evidence indicating that the female estrus cycle significantly influences ultrasonic vocalization (Matochik et al., 1992). Adhering to controls for the estrus cycle would require a greater number of female subjects than males, which would not only increase animal suffering but also escalate the demands of human labor and financial costs.

> *Firstly, the authors argue that the shift to higher-frequency aversive calls is due to an increase in arousal (caused by the animals having received multiple aversive foot shocks towards the end of the protocols). However, it cannot be ruled out that this shift would be due to factors such as the passage of time and increase in fatigue of the animals as they make vocalizations (and other responses) for extended periods of time. In fact the gradual frequency increase reported for 22 kHz calls and the drop in 44 kHz calls the next day in testing is in line with this.*

Answer: We would like to point out that the “increased-arousal” hypothesis, declared in the manuscript, is only a hypothesis – as reflected by the wording used. However, we changed the beginning of the sentence in question from “It could be argued” to “We would like to propose a hypothesis” to emphasize the speculative aspect of the proposed explanation behind the increase of 44-kHz ultrasonic emissions.

Also, we do agree that other factors could contribute to the increased emission of 44kHz calls. These factors could include: heightened fear, stress/anxiety, annoyance/anger, disgust/boredom, grief/sadness, despair/helplessness, and weariness/fatigue. We are listing these potential factors in the discussion. Also, we added: “It is not possible, at this stage, to determine which factors played a decisive role. Please note that the potential contribution of these factors is not mutually exclusive”. However, we propose a list of arguments supporting the idea that 44-kHz vocalizations communicate an increased negative emotional state. Among these arguments were the conclusions drawn from additional analyses – mostly inspired by the fatigue hypothesis proposed by the Reviewer #1. In particular, we investigated changes in the sound mean power and duration of 22-kHz and 44-kHz calls. Specifically, we showed that the mean power of 44-kHz vocalizations did not change, and was higher than that of 22-kHz vocalizations (Fig. 1S2EF).

Finally, the Reviewer #1 listed “the gradual frequency increase reported for 22 kHz calls and the drop in 44 kHz calls the next day” as arguments for the fatigue hypothesis. We do not agree that the “increase” should be interpreted as a sign of fatigue [Producing and maintaining higher frequency calls require greater effort from the vocalizer, on which we elaborated in the manuscript], also we are not sure what “drop in 44 kHz calls” the Reviewer is referring to [We assume it refers to less 44-kHz calls during testing vs. training; we suppose that the levels of arousal are lower in the test due to shorter session time and lack of shocks, which additionally contributes to fear extinction].

> *Secondly, regarding the analysis where calls were sorted using DBSCAN based on peak frequency and duration, it is not surprising that the calls cluster based on frequency and duration, i.e. the features that are used to define the 44 kHz calls in the first place. Thus presenting this clustering as evidence of them being truly distinct call types comes across as a circular argument.*

Answer: The DBSCAN sorting results were to convey that when changing the clustering ε value, the degree of cluster separation, the 44-kHz vocalizations remained distinct from the 22-kHz and various short-call clusters that merged. In other words: 44-kHz calls remained separate from long 22-kHz, short 22-kHz and 50-kHz vocalizations, which all consolidated into one common cluster. As a result, in this mathematical analysis, 44-kHz vocalizations remained distinct without applying human biases. Additionally, frequency and duration are the two most common features used to define all types of calls (Barker et al., 2010; Silkstone & Brudzynski, 2019a, 2019b; Willey & Spear, 2013). In summary, we did not expect the analysis to isolate out the 44-kHz calls, and we were surprised by this result.

> *The sparsity of calls in the 30-40 kHz range (shown in the individual animal panels in Figure 2C) could in theory be explained by some bioacoustics properties of rat vocal cords, without necessarily the calls below and above that range being ethologically distinct.*

Answer: We respectfully disagree with the argument regarding sparsity. It is important to note that, during prolonged fear conditioning experiments, we observed an increased incidence of 44-kHz calls (Fig. 1E-G) of up to >19% (Fig. 1S2AB) of the total ultrasonic vocalizations during specific inter-trial intervals. Also, it is possible that in observed experimental circumstances almost every fifth call could be attributed to the vocal apparatus as an artifact of its functioning (assuming we are interpreting the Reviewer's argument correctly). While we do not believe this to be the case, we acknowledge the importance of considering such a hypothesis.

> *The behavioral response to call playback is intriguing, although again more in line with the hypothesis that these are not a distinct type of call but merely represent expected variation in vocalization parameters. Across the board animals respond rather similarly to hearing 22 kHz calls as they do to hearing 44 kHz calls, with occasional shifts of 44 kHz call responses to an intermediate between appetitive and aversive calls. This does raise interesting questions about how, ethologically, animals may interpret such variation and integrate this interpretation in their responses. However, the categorical approach employed here does not address these questions fully.*

Answer: We are unsure of the Reviewer's critique in this paragraph and will attempt to address it to the best of our understanding. Our finding of up to >19% of long seemingly aversive, 44-kHz calls, at a frequency in the define appetitive ultrasonic range (usually >32 kHz) is unexpected rather than "expected". We would agree that aversive call variation is expected, but not in the appetitive frequency range.

Kindly note the findings by Saito et al. (2019), which claim that frequency band plays the main role in rat ultrasonic perception. It is possible that the higher peak frequency of 44kHz calls may be a strong factor in their perception by rats, which is, however, modified by the longer duration and the lack of modulation.

Also, from our experience, it is quite challenging to demonstrate different behavioral responses of naïve rats to pre-recorded 22-kHz (aversive) vs. 50-kHz (appetitive) vocalizations. Therefore, to demonstrate a difference in response to two distinct, potentially aversive, calls, i.e., 22-kHz vs. 44-kHz calls, to be even more difficult (as to our knowledge, a comparable experiment between short vs. long 22-kHz ultrasonic vocalizations, has not been done before).

Therefore, we do not take lightly the surprising and interesting finding that "animals respond rather similarly to hearing 22 kHz calls as they do to hearing 44 kHz calls, with occasional shifts of 44 kHz call responses to an intermediate between appetitive and aversive calls". We would rather put this description in analogous words: "the rats responded similarly to hearing 44-kHz calls as they did to hearing aversive 22-kHz calls, especially regarding heartrate change, despite the 44-kHz calls occupying the frequency band of appetitive 50-kHz vocalizations" and "other responses to 44-kHz calls were intermediate, they fell between response levels to appetitive vs. aversive playback" – which we added to the Discussion.

Finally, we acknowledge that our findings do not present a finite and complete picture of the discussed aspects of behavioral responses to the presented ultrasonic stimuli (44-kHz vocalizations). Therefore, we have incorporated the Reviewer's suggestion in the discussion. The added sentence reads: "Overall, these initial results raise further questions about how, ethologically, animals may interpret the variation in hearing 22-kHz vs. 44-kHz calls and integrate this interpretation in their responses."

> *In sum, rather than describing the 44kHz long calls as a new call type, it may be more accurate to say that sometimes aversive calls can occur at frequencies above 22 kHz. Individual and situational variability in vocalization parameters seems to be expected, much more so than all members of a species strictly adhering to extremely non-variable behavioral outputs.*

Answer: The surprising fact that there are presumably aversive calls that are beyond the commonly applied thresholds, i.e. >32 kHz, while sharing some characteristics with 22-kHz calls, is the main finding of the current publication. Whether they be finally assigned as a new type, subtype, i.e. a separate category or become a supergroup of aversive calls with 22-kHz vocalizations is of secondary importance to be discussed with other researchers of the field of study.

However, we would argue – by showing a comparison – that 22-kHz calls occur at durations of <300 ms and also >300 ms, and are, usually, referred to in literature as short and long 22-kHz vocalizations, respectively (not introduced with a description that “sometimes 22kHz calls can occur at durations below 300 ms”). These are then regarded and investigated as separate groups or classes usually referred to as two different “types” (e.g., Barker et al., 2010) or “subtypes” (e.g., Brudzynski, 2015). Analogously, 44-kHz vocalizations can also be regarded as a separate type or a subtype of 22-kHz calls. The problem with the latter is that 22-kHz vocalizations are traditionally and predominantly defined by 18–32 kHz frequency bandwidth (Araya et al., 2020; Barroso et al., 2019; Browning et al., 2011; Brudzynski et al., 1993; Hinchcliffe et al., 2022; Willey & Spear, 2013).

> ***Reviewer #2 (Public Review):***
>
> *Olszyński et al. claim that they identified a "new-type" ultrasonic vocalization around 44 kHz that occurs in response to prolonged fear conditioning (using foot-shocks of relatively high intensity, i.e. 1 mA) in rats. Typically, negative 22-kHz calls and positive 50-kHz calls are distinguished in rats, commonly by using a frequency threshold of 30 or 32 kHz. Olszyński et al. now observed so-called "44-kHz" calls in a substantial number of subjects exposed to 10 tone-shock pairings, yet call emission rate was low (according to Fig. 1G around 15%, according to the result text around 7.5%).*

Answer: We are thankful for praising the strengths. Please note Figure 1G referred to 10-trial Wistar rats during delay fear conditioning session in which 44-kHz constituted 14.1% of ultrasonic vocalizations. The 7.5% number in results refers to the total of vocalizations analyzed across all animal groups used in fear conditioning experiments. These values have been updated in the current version of the manuscript. Also, please note – 44-kHz calls constituted up to 19.4% of calls, on average, in one of the ITI during fear conditioning session. However, the prevalence of aversive calls and of 44-kHz vocalizations in particular varied. It varied between individual rats; we added the text: “for n = 3 rats, 44-kHz vocalizations accounted for >95% of all calls during at least one ITI (e.g., 140 of total 142, 222 of 231, and 263 of 265 tallied 44-kHz calls), and in n = 9 rats, 44-kHz vocalizations constituted >50% of calls in more than one ITI.” See also further for the description of the array of experiments analyzed and the prevalence/percentage of 44-kHz calls encountered (Tab. 1, Fig. 1S3).

> *Weaknesses: I see a number of major weaknesses.*

While the descriptive approach applied is useful, the findings have only focused importance and scope, given the low prevalence of "44 kHz" calls and limited attempts made to systematically manipulate factors that lead to their emission. In fact, the data presented appear to be derived from reanalyses of previously conducted studies in most cases and the main claims are only partially supported. While reading the manuscript, I got the impression that the data presented here are linked to two or three previously published studies (Olszyński et al., 2020, 2021, 2023). This is important to emphasize for two reasons:

> *(1) It is often difficult (if not impossible) to link the reported data to the different experiments conducted before (and the individual experimental conditions therein). While reanalyzing previously collected data can lead to important insight, it is important to describe in a clear and transparent manner what data were obtained in what experiment (and more specifically, in what exact experimental condition) to allow appropriate interpretation of the data. For example, it is said that in the "trace fear*

> *conditioning experiment" both single- and grouphoused rats were included, yet I was not able to tell what data were obtained in single- versus group-housed rats. This may sound like a side aspect, however, in my view this is not a side aspect given the fact that ultrasonic vocalizations are used for communication and communication is affected by the social housing conditions.*

Answer: Preparing the current manuscript, we indeed used data collected during fear conditioning experiments which were described previously (Olszyński et al., 2021; Olszyński et al., 2022). Please note, however, that vocalization behavior during the fear conditioning itself was not the main subject of these publications. Our previous publications (Olszyński et al., 2020; Olszyński et al., 2021; Olszyński et al., 2022) present primarily ultrasonic-vocalization data from playback-part of experiments whereas here we analyze recordings obtained during fear conditioning experiments, thus we are analyzing new parts, i.e., not yet analyzed, of previously published studies. Also, we have performed additional experiments.

In the first version of the current manuscript, we did not attempt to demonstrate exactly which calls were recorded in which conditions as the focus was to demonstrate that 44-kHz calls were emitted in several different fear-conditioning experiments. Also, as the experiments were not performed simultaneously and are results from different experimental situations, we would prefer to not compare these results directly.

However, in the current version of the manuscript, we have introduced an additional reference system, based on Tab. 1, to more clearly indicate which rats have been employed in each analysis, e.g. the group of “Wistar rats that undergone 10 trials of fear conditioning” are described as “Tab. 1/Exp. 1-3/#2,4,8,13; n = 46”, i.e., these are the rats listed in rows 2, 4, 8, and 13 of Tab. 1.

We have also tried to unify the analyses, in terms of rats used, as much as possible. Finally, we have also introduced Fig. 1S3 to demonstrate the prevalence of 44-kHz calls in all experiments analyzed with the note that “the experiments were not performed in parallel”.

Regarding the Reviewer’s concerns about analyzing single- and pair-housed rats together. We have examined ultrasonic vocalizations emitted and freezing behavior in these two groups.

• Ultrasonic vocalizations; when comparing the number of vocalizations, their duration, peak frequency and latency to first occurrence, equally for all types of calls and divided into types (short 22-kHz, long 22-kHz, 44-kHz, 50-kHz), the only difference was observed in peak frequency in 50-kHz vocalizations (50.7 ± 2.8 kHz for paired vs. 61.8 ± 3.1 kHz for single rats; $p = 0.0280$, Mann-Whitney). Since 50-kHz calls are not the subject of the current publication, we did not investigate this difference further. Also, this difference was not observed during playback experiments (Olszyński et al., 2020, Tab. 1).

• Freezing. There were no differences between single- and pair-housed groups in freezing behavior, both in the time before first shock presentation and during fear conditioning training (Mann-Whitney).

In summary, since the two groups did not differ in relevant ultrasonic features and freezing, we decided to present the results obtained from these rats together. However, we agree with the Reviewer, and it is possible that social housing conditions may in fact affect the emission of 44-kHz vocalizations, which could be a subject of another project – involving, e.g., larger experimental groups observed under hypothesis-oriented and defined conditions.

> *(2) In at least two of the previously published manuscripts (Olszyński et al., 2021, 2023), emission of ultrasonic vocalizations was analyzed (Figure S1 in Olszyński et al., 2021, and Fig. 1 in Olszyński et al., 2023). This includes detailed spectrographic analyses covering the frequency range between 20 and 100 kHz, i.e. including the frequency range, where*

> *the "newtype" ultrasonic vocalization, now named "44 kHz" call, occurs, as reflected in the examples provided in Fig. 1 of Olszyński et al. (2023). In the materials and methods there, it was said: "USV were assigned to one of three categories: 50-kHz (mean peak frequency, MPF >32 kHz), short 22-kHz (MPF of 18-32 kHz, <0.3 s duration), long 22-kHz (MPF of 18-32 kHz, >0.3 s duration)". Does that mean that the "44 kHz" calls were previously included in the count for 50-kHz calls? Or were 44 kHz calls (intentionally?) left out? What does that mean for the interpretation of the previously published data? What does that mean for the current data set? In my view, there is a lack of transparency here.*

Answer: As mentioned above, we indeed used data collected during fear conditioning experiments which were described previously (Olszyński et al., 2021; Olszyński et al., 2022). However, in these publications, ultrasonic vocalizations emitted during playback experiments were the main subject, while the ultrasonic calls emitted during fear conditioning (performed before the playback) were only analyzed in a preliminary way. As a result, the 44-kHz vocalizations analyzed in the current manuscript were not included in the previous analyses. In particular, in Olszyński et al. (2021), we counted the overall number of ultrasonic vocalizations before fear conditioning session to determine the basal ultrasonic emissions (Fig. S1). Then, our next article (Olszyński et al., 2022), we analyzed again the number of all ultrasonic vocalizations before fear conditioning (Fig. S1) and restricted the analysis of vocalizations during fear conditioning to 22-kHz calls (Tab. S1 and S2).

Also, we re-reviewed all the data used in our previous playback publications. Overall, 44-kHz calls were extremely rare in playback parts of the experiments. There were no 44-kHz calls in the playback data used in Olszyński et al. (2022) and Olszyński et al. (2020). In Olszyński et al. (2021), one rat produced eight 44-kHz calls. These 44-kHz calls constituted 0.03% of all vocalizations analyzed in the experiment (8/24888) and were included in the total number of calls analyzed (but not in the 50-kHz group), they were not described in further detail in that publication.

> *Moreover, whether the newly identified call type is indeed novel is questionable, as also mentioned by the authors in their discussion section. While they wrote in the introduction that "high-pitch (>32 kHz), long and monotonous ultrasonic vocalizations have not yet been described", they wrote in the discussion that "long (or not that long (Biały et al., 2019)), frequency-stable high-pitch vocalizations have been reported before (e.g. Sales, 1979; Shimoju et al., 2020), notably as caused by intense cholinergic stimulation (Brudzynski and Bihari, 1990) or higher shock-dose fear conditioning (Wöhr et al., 2005)" (and I wish to add that to my knowledge this list provided by the authors is incomplete). Therefore, I believe, the strong claims made in abstract ("we are the first to describe a new-type..."), introduction ("have not yet been described"), and results ("new calls") are not justified.*

Answer: We would argue that 44-kHz vocalizations were indeed reported but not described. As far as we are concerned, an in-depth analysis of the properties and experimental circumstance of emission of long, high-frequency calls has not yet been performed. These researchers have observed, at least to a degree, similar calls to the ones we observed – as we mentioned in the discussion section. However, since these reported 44-kHz vocalizations were not fully described, we can only guess that they may be similar to ours. We speculate that perhaps like us, these researchers unknowingly recorded 44-kHz calls in their experiments and may also be able to describe them more extensively when re-analyzing their data as we have done here.

Possibly, it was difficult to find reports on vocalizations, similar to the 44-kHz calls that we observed, because of the canonical and accepted definitions of ultrasonic vocalization types. Biały et al. (2019) allocated them as a part of 22-kHz group, perhaps because their calls were often of a step variation having both low and high components. Shimoju et al. (2020) grouped them along with 50-kHz vocalizations because they appeared during stroking rats held vertically; this procedure was compared to tickling which usually elicits appetitive calls.

The Reviewer #2 states there are other publications to complete the list. We are aware of other articles authored by the same team as Shimoju et al. (2020) with different first authors. However, they are reporting similar findings to the cited article. Otherwise, we would gladly cite a more complete list of publications showing atypical, long, monotonous highfrequency vocalizations, similar to those observed in our experiments. Therefore, we would argue that ultrasonic vocalizations which were long, flat, high in frequency, and repeatedly occurring in a defined behavioral situation, have not been reported before. However, concerning the strong claims of novelty of our finding, we toned them down where we found this was warranted.

> *In general, the manuscript is not well written/ not well organized, the description of the methods is insufficient, and it is often difficult (if not impossible) to link the reported data to the experiments/ experimental conditions described in the materials and methods section.*

Answer: The description of the methods has been adjusted and expanded. We added the requested link to each particular experiment as a formula “Tab. 1/Exp. nos./# nos.” which shows, each time, which experiments and experimental groups were analyzed. The list of the experiments and groups is found in the Tab. 1.

> *For example, I miss a clear presentation of basic information: 1) How many rats emitted "44 kHz" calls (in total, per experiment, and importantly, also per experimental condition, i.e. single- versus group-housed)?*

Answer: We now clearly show which experiments were performed and how many animals were tested in each condition (Tab. 1), while the prevalence of 44-kHz calls amongst experimental conditions and animal groups is shown in Fig. 1S3. Also, we included information regarding the number of animals and treatment of each group of rats when reporting results. For example, we are stating that:

(1a) “53 of all 84 conditioned Wistar rats (Tab. 1/Exp. 1-3/#2,4,6-8,13, Figs 1B, 1E, 1S1BC) displayed” 44-kHz vocalizations – as a general assessment; these numbers are different from those in the first version of the Ms, when we are mentioning Wistar rats conditioned 6 or 10 times only.

(1b) “From this group of rats (n = 46), n = 41 (89.1%) emitted long 22-kHz calls, and 32 of them (69.6%) emitted 44-kHz calls” – this time referring only to 10-times conditioned Wistar rats as the biggest group that could be analyzed together (Figs 1F, 1G, 1S2A).

(1c) “for n = 3 rats, 44-kHz vocalizations accounted for >95% of all calls during at least one ITI (e.g., 140 of total 142, 222 of 231, and 263 of 265 tallied 44-kHz calls), and in n = 9 rats, 44kHz vocalizations constituted >50% of calls in more than one ITI.”

> *(2) Out of the ones emitting "44 kHz" calls, what was the prevalence of "44 kHz" calls (relative to 22- and 50-kHz calls, e.g. shown as percentage)?*

Answer: The prevalence of 44-kHz vocalizations in all investigated experiments and groups is shown in Fig. 1S3CD. Also, more information regarding the percentage of 44-kHz calls was demonstrated in Fig. 1S2AB where we calculated the distribution of 44-kHz calls to 22-kHz calls in Wistar rats, in 10-trial fear conditioning, across the length of the session.

Additionally, the values are listed in the sentence regarding all Wistar rats which underwent 10 trials of fear conditioning: “these vocalizations were less frequent following the first trial

(1.2 ± 0.4% of all calls), and increased in subsequent trials, particularly after the 5th (8.8 ± 2.8%), through the 9th (19.4 ± 5.5%, the highest value), and the 10th (15.5 ± 4.9%) trials, where 44-kHz calls gradually replaced 22-kHz vocalizations in some rats (Fig. 1F, 1S2B, Video 1; comp Fig. 1D vs. 1E).”

> *(3) How did this ratio differ between experiments and experimental conditions?*

Answer: The prevalence of 44-kHz vocalizations in all experimental conditions is shown in Fig. 1S3. However, the direct comparison of results obtained in different conditions was not the goal of the present work. Also, we would argue, that such direct comparisons of results of different experiments would not be allowed. These experiments were done with different groups of animals, at different times, with different timetables of experimental manipulations.

However, we are comfortable to state that:

- There were more 44-kHz vocalizations during fear conditioning training than testing in all fear-conditioned Wistar rats;
- We observed more 44-kHz vocalizations in Wistar rats compared to SHR.

> *(4) Was there a link to freezing? Freezing was apparently analyzed before (Olszyński et al., 2021, 2023) and it would be important to see whether there is a correlation between "44-kHz" calls and freezing. Moreover, it would be important to know what behavior the rats are displaying while such "44-kHz" calls are emitted? (Note: Even not all 22-kHz calls are synced to freezing.) All this could help to substantiate the currently highly speculative claims made in the discussion section ("frequency increases with an increase in arousal" and "it could be argued that our prolonged fear conditioning increased the arousal of the rats with no change in the valence of the aversive stimuli"). Such more detailed analyses are also important to rule out the possibility that the "new-type" ultrasonic vocalization, the so-called "44 kHz" call, is simply associated with movement/ thorax compression.*

Answer: We analyzed freezing behavior and its association with ultrasonic emissions. The emission of 44-kHz vocalizations was associated with freezing. The results are now described and presented in the manuscript, i.e., Tab. 2, its legend and the description in Results: “Freezing during the bins of 22-kHz calls only ($p < 0.0001$, for both groups) and during 44-kHz calls only bins ($p = 0.0003$) was higher than during the first 5 min baseline freezing levels of the session. Also, the freezing associated with emissions of 44-kHz calls only was higher than during bins with no ultrasonic vocalizations ($p = 0.0353$), and it was also 9.9 percentage points higher than during time bins with only long 22-kHz vocalizations, but the difference was not significant ($p = 0.1907$; all Wilcoxon)” and “To further investigate this potential difference, we measured freezing during the emission of randomly selected single 44-kHz and 22-kHz vocalizations. The minimal freezing behavior detection window was reduced to compensate for the higher resolution of the measurements (3, 5, 10, or 15 video frames were used). There was no difference in freezing during the emission of 44-kHz vs. 22-kHz vocalizations for ≥150ms-long calls (3 frames, $p = 0.2054$) and for ≥500-ms-long calls (5 frames, $p = 0.2404$; 10 frames, $p = 0.4498$; 15 frames, $p = 0.7776$; all Wilcoxon, Tab. 2B).”

Please note, that the general observation that "frequency increases with an increase in arousal" is not our claim but a general rule derived from large body of observations and proposed by the others (Briefer et al., 2012); we changed the wording of this statement to: “frequency usually increases with an increase in arousal (Briefer et al., 2012)”.

> *The figures currently included are purely descriptive in most cases - and many of them are just examples of individual rats (e.g. majority of Fig. 1, all of Fig. 2 to my understanding, with the exception of the time course, which in case of D is only a subset of rats ("only rats that emitted 44-kHz calls in at least seven ITI are plotted" - is there any rationale for this criterion?)), or, in fact, just representative spectrograms of calls (all of Fig. 3, with the exception of G, all of Fig. 4).*

Answer: Please note, the former figures 2, 4, 6, and 8 have been now moved to supplementary figures 1S1, 2S1, 3S1, and 4S1 – to better organize the presentation of data. Figures 1, 3, 5, 7 are now 1, 2, 3, 4 respectively. In regards to presenting data from individual rats, this was to show the general patterns of ultrasonic-calls distributions observed. Showing the full data set as seen in Fig. 5A (now Fig. 3A) would obscure the readability of the graph without using mathematical clustering techniques such as DBSCAN.

Concerning the Reviewer's #2 question regarding the criterion of "minimum seven ITI", we selected the highest vocalizers by taking animals above the 75th percentile of the number of ITI with 44-kHz calls. However, in the current version of the manuscript, we decided to omit this part of the analysis and the accompanying part of the figure, since it did not provide any additional informative value (apart from employing questionable criterion).

> *Moreover, the differences between Fig. 5 and Fig. 6 are not clear to me. It seems Fig. 5B is included three times - what is the benefit of including the same figure three times?*

Answer: We hope that designating Fig. 6 as supplementary to Fig. 5 (now Figs 3S1 and 3, respectively) will make interpreting them more streamlined. Fig. 6A (now Fig. 3S1A) is a more detailed look on information presented in Fig. 5B (now Fig. 3B) with spectrogram images of ultrasonic vocalizations from different areas of the plot. Also, Fig. 3B (former Fig. 5B) was removed from Fig. 3S1B (former Fig. 6B).

> *A systematic comparison of experimental conditions is limited to Fig. 7 and Fig. 8, the figures depicting the playback results (which led to the conclusion that "the responses to 44-kHz aversive calls presented from the speaker were either similar to 22-kHz vocalizations or in between responses to 22-kHz and 50-kHz playbacks", although it remains unclear to me why differences were seen b e f o r e the experimental manipulation, i.e. the different playback types in Fig. 8B).*

Answer: There were indeed instances of such before-differences. Such differences were observed in our previous studies (Olszyński et al., 2020, Tabs S9-12; Olszyński et al., 2021, Tabs S7; Olszyński et al., 2022, Tabs S4, S9, S13, S17, S18) and were most likely due to analyzing multiple comparisons. However, we think that the carry-over effect, mentioned by the Reviewer #2 (see below), also played a role.

> *Related to that, I miss a clear presentation of relevant methodological aspects: 1) Why were some rats single-housed but not the others?*

Answer: As stated before, data were collected from our previous experiments and the observation of 44-kHz vocalizations in fear conditioning was an emergent discovery as we decided to analyze ultrasonic recordings from fear conditioning procedures. Single-housed animals were part of our experiment comparing fear conditioning and social situation on the perception of ultrasonic playback as described in Olszyński et al. (2020). Aside from this experiment, all other rats were housed in pairs.

> *(2) Is the experimental design of the playback study not confounded? It is said that "one group (n = 13) heard 50-kHz appetitive vocalization playback while the other (n = 16) 22-kHz and 44kHz aversive calls". How can one compare "44 kHz" calls to 22- and 50-kHz calls when "44 kHz" calls are presented together with 22-kHz calls but not 50-kHz calls? What about carry-over effects? Hearing one type of call most likely affects the response to the other type of call. It appears likely that rats are a bit more anxious after hearing aversive 22-kHz calls, for example. Therefore, it would not be very surprising to see that the response to "44 kHz" calls is more similar to 22-kHz calls than 50-kHz calls.*
>
> *Of note, in case of the other playback experiment it is just said that rats "received appetitive and aversive ultrasonic vocalization playback" but it remains unclear whether "44 kHz" calls are seen as appetitive or aversive. Later it says that "rats were presented with two 10-s-long playback sets of either 22-kHz or 44-kHz calls, followed by one 50-kHz modulated call 10-s set and another two playback sets of either 44-kHz or 22-kHz calls not previously heard" (and wonder what data set was included in the figures and how - pooled?). Again, I am worried about carry-over effects here. This does not seem to be an experimental design that allows to compare the response to the three main call types in an unbiased manner.*

Answer: We apologize for being confounding and brief in our original description of the playback experiments. We wanted to avoid confusion associated with including several additional playback signals (please note some are not related to the current comparisons and include different 50-kHz ultrasonic subtypes and two different subtypes of short 22-kHz calls). We lengthened the description of these playback experiments in the current version.

In general, including more than one type of ultrasonic calls as playback has a risk of a carry-over effect as well as a habituation effect (the responses become weak). However, it greatly reduces the number of required animals. Finally, regarding the first experiment, we chose 3 playbacks to compare the rats' reactions, as this was the most conservative choice we thought of.

We would like to highlight that we wanted to compare specifically the rats' responses to 22-kHz vs. 44-kHz playback (as well as the effects of playback of different subtypes 50-kHz calls, which is not the subject of the current work). Therefore, we would argue, that the design of both experiments is actually unbiased regarding this key comparison (responses to 22-kHz vs. 44-kHz playback). In both experiments, 22-kHz and 44-kHz playbacks were included in the same sequences of stimuli and counterbalanced regarding their order (i.e., taking into account possible carry-over effects), and presented to the same rats. We regarded the group of rats that heard 50-kHz recordings as a baseline/control, since we know from previous playback studies what reactions to expect from rats exposed to these vocalizations (and 22-kHz playback), while in the second experiment, we reduced the 50-kHz playback to one set in order to minimize possible habituation to multiple playbacks.

We agree that the design of both experiments does not allow for full comparison of the effects of aversive playbacks to 50-kHz playback. Also, we agree that some carry-over effects could play a role. It was mentioned in the discussion: "Please factor in potential carryover effects (resulting from hearing playbacks of the same valence in a row) in the differences between responses to 50-kHz vs. 22/44-kHz playbacks, especially, those observed before the signal (Fig. 4AB)." However, we would still argue that the observed lack of difference in heartrate response (Fig. 4A) and the differences regarding the number of 50-kHz calls emitted (e.g., Fig. 4S1F) are void of the constraints raised by the Reviewer #2.

We acknowledge that our studies do not give a complete picture of 44-kHz ultrasonic perception in relation to other ultrasonic bands and, given the possibility, we would like to perform more in-depth and focused experiments to study this aspect of 44-kHz calls in the future.

Finally, regarding the second experiment, the description of the rats now includes that they "received 22-kHz, 44-kHz, and 50-kHz ultrasonic vocalization playback", while the description of the experiment itself includes: "Responses to the pairs of playback sets were averaged".

> *Of note, what exactly is meant by "control rats" in the context of fear conditioning is also not clear to me. One can think of many different controls in a fear conditioning experiment.*
>
> *More concrete information is needed.*

Answer: This information was included in our previous publications. However, it was now provided in the method section of the current version of the manuscript. In general, control rats were subjected to the same procedures but did not receive electric shocks.

Literature included in the answers

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> ***Recommendations For The Authors:***
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> ***Reviewer #1 (Recommendations For The Authors):***
>
> *Additional considerations:*
>
> *The discussion of the "perfect fifth" and the proposition that this observation could be evidence of an evolutionary mechanism underlying it is rather far-fetched, especially for*

> *being presented in the Results section (with no supporting non-anecdotal evidence).*

Answer: We agree with the Reviewer #1. The text was modified, the word “evolutionary” was deleted. Instead, we expended on the possible reason for prevalence of the perfect fifth in the current version of the manuscript; we added that the prevalence of the perfect fifth: “could be explained by the observation that all physical objects capable of producing tonal sounds generate harmonic vibrations, the most prominent being the octave, perfect fifth, and major third (Christensen, 1993, discussed in Bowling and Purves, 2015).”

> *It is not clear why Sprague-Dawleys were used as "receivers" in the playback experiment, when presumably the calls were recorded from Wistars and SHRs. While this does not critically impact the conclusions, within the species rats should be able to respond appropriately to calls made by rats of different genetic backgrounds, it adds an unnecessary source of variance.*

Answer: Sprague-Dawley rats were used to test another normotensive strain of rats. Regarding the Reviewer’s main point – we beg to differ as we think that it is worth testing playback stimuli in different strains. Diverging the stimuli between different rat strains would add unnecessary variance and it seemed logical to use the same recordings to test effects in different strains. Please note that finally, in spite of this additional variance, the results of both playback experiments are, in general, similar – which may point to a universal effect of 44-kHz playback across rat strains.

> *It is pertinent to note that for the trace fear conditioning experiment, the rats had previously been exposed to a vocalization playback experiment. While such a pre-exposure is unlikely to be a very strong stressor, the possibility for it to influence the vocal behaviors of these rats in later experiments cannot be ruled out. It is also not clear what the control rats in this experiment experienced (home cage only?), nor what they were used for in analyses.*

Answer: In the current version of the manuscript, we have described in greater detail all the experiments performed and analyzed. We would like to emphasize that both delay and trace fear conditioning experiments with radiotelemetric transmitters were not performed specifically to elicit any particular response during fear conditioning, rather that our observation of 44-kHz vocalizations emerged as a result of re-examining the audio recordings. As a result, this work summarizes our observations of 44-kHz calls from several different experiments. It is relevant to note, that 44-kHz vocalizations were observed “in rats which were exposed to vocalization playback experiment”, in rats before the playback experiments as well as in naïve rats, without transmitters implemented, trained in fear conditioning (Tab. 1/Exp. 1-3).

Our main message is that 44-kHz vocalizations were present in several experiments, with different conditions and subjects, while we are not attempting to compare in detail the results across the different experiments. In other words, we agree that pre-exposure to playback (and even more likely – transmitters implantation) could influence, but are not necessary, for 44-kHz ultrasonic emissions by the rats. To demonstrate this, we added a prolonged fear conditioning group with naïve Wistar rats (Exp. 3) to verify the emission of 44kHz calls in the absence of those experimental factors.

We modified the methods section to clarify the circumstances under which these discoveries were made, such as including the information regarding the control rats in trace fear conditioning. In particular we mention that: “Control rats were subjected to the exact same procedures but did not receive the electric shock at the end of trace periods”.

> *For Figure 1A-E, only example call distributions from individual rats are shown. It would perhaps be more informative to see the full data set displayed in this manner, with color/shape codes distinguishing individuals if desired.*

Answer: Please note the Fig. 1S1 shows more examples of ultrasonic call distribution. Showing all the data would make it more difficult to read and interpret. The problem is partly amended in Fig. 3A.

> *It is not clear what is presented in Figure 2D vs. E, i.e. panel D is shown only for "selected rats" but the legend does not clarify how and why these rats were selected. It is also not clear why the legend reports p-values for both Friedman and Wilcoxon tests; the latter is appropriate for paired data which seems to be the case when the question is whether the call peak frequency alters across time, but the Friedman assumes non-paired input data.*

Answer: The question refers to the current Fig. 1S2C panel (former Fig. 2E panel) and the former Fig. 2D panel. The latter was not included in the current version of the manuscript, since both reviewers opposed the presentation of “selected rats” only (see above). The full description of the Fig. 1S2C panel is now in the results section together with p-values for Friedman and Wilcoxon test. We used the latter to investigate the difference between the first and the last ITI (selected paired data), while the Friedman to investigate the presence of change within the chain of ten ITI – since it is a suitable test for a difference between two or more paired samples.

> ***Reviewer #2 (Recommendations For The Authors):***
>
> *The weaknesses listed in the public review need to be addressed.*

Answer: We have done our best to address the weaknesses.

> *Notes: 1) Page and line numbers would have been useful.*

Answer: We are including a separate manuscript version with page and line numbers.

.(2) English language needs to be improved.

Answer: The text has been checked by two native English speakers (one with a scientific background). Both only identified minor changes to improve the text which we applied.

> *(3) I am a bit unsure whether the comment about the Star Wars movie (1997) and the Game of Thrones series (2011) is supposed to be a joke.*

Answer: These are indeed two genuine examples of the perfect fifth in human music that we hope are easily recognizable and familiar to readers. Parts of the same examples of the perfect fifth can also heard in the rat voice files provided.